

Scalable Share Calibration for Factor Multinomial Probit Models

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Abstract

We give an algorithm that inverts observed shares to utilities in multinomial probit models with many alternatives, when the covariance is low-rank-plus-diagonal and supplied. Conditioning on the factors makes the alternatives independent, so one shared survival field prices all N alternatives in a single pass over Q factor nodes and L grid points, at $O(QNL)$ cost. The same field yields resampling-free Jacobian–vector products, and share inversion at $N = 1000$ takes three seconds in the compiled implementation at rank two. Ten thousand alternatives stay under a minute. In convergence comparisons, the computed shares carry log-share errors of a few parts in a hundred thousand, hundred-to-one shares included. The established high-precision evaluators price one orthant at a time. The nearest factor-aware competitor computes the same share vector roughly $140\times$ slower at matched accuracy at $N = 1000$, and the gap widens linearly in N .

Keywords: multinomial probit; discrete choice; numerical quadrature; Jacobian–vector product; share inversion; factor model

1 Introduction

This paper solves a calibration problem. The data are N shares (choice frequencies, market shares, win probabilities), one per alternative.¹ The model is multinomial probit: jointly Gaussian utilities whose covariance is low-rank-plus-diagonal,

$$\Sigma = VV^\top + \text{diag}(D),$$

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¹We call the choice probabilities p_i *shares* throughout. The racing origin (Cotton, 2021a) called them *state prices*, borrowing prediction-market and finance usage. In the probit choice setting *shares*, as in market shares, is the more natural term. *Probability* would also be imprecise on the discrete lattice used later, where ties have positive probability and a dead heat splits the payout. The lattice objects really are state prices.

a race in which each performance is idiosyncratic but also influenced by a small number of factors common to all runners. The factor structure (V, D) is fixed and supplied. The problem is to recover the utilities at which the model reproduces the shares, together with the derivatives and removal counterfactuals that make the calibrated model usable, at N in the thousands.²

The logit version of this inversion is daily routine. Demand estimation in the Berry–Levinsohn–Pakes (BLP) tradition begins by inverting observed market shares to mean utilities, and runs on the closed forms and contraction mapping that the Gumbel distribution supplies (Berry, 1994; Berry et al., 1995; Conlon and Gortmaker, 2020). The probit version is treated as out of reach. The record is explicit, and we quote it before removing the barrier for the covariance class above.

The model itself is the oldest in choice theory. Thurstone’s 1927 law of comparative judgment posited jointly Gaussian utilities with unrestricted correlation (Thurstone, 1927). Econometrics reintroduced the same object as the conditional, or multinomial, probit model (MNP) (Hausman and Wise, 1978). It re-emerges wherever latent performance is Gaussian, from bioassay and quantitative genetics to item response theory, signal detection, credit risk, skill rating, and the final layer of heteroskedastic image classifiers (Collier et al., 2021). Section 9 collects these fields. The computation below applies to all of them. Any empirical argument for one race over another on real data is deferred to a companion survey (Cotton, 2026). Here the focus is the computation, and the record of its difficulty, which is clearest in psychometrics and econometrics.

Psychometrics stated the backward problem and abandoned it. Torgerson’s text records the degrees-of-freedom count, n observed frequencies against n scale values, so “we ought, therefore, to be able to fit perfectly”. It also records the retreat that followed. Every working method from Thurstone forward, down to modern Thurstonian rating engines such as TrueSkill, reduces the N -way contest to pairwise events (Torgerson, 1958; Herbrich et al., 2006).

Econometrics kept the N -way event and changed the distribution. A major reason for logit’s dominance is brutally practical. Logit models can be computed. Probit probabilities are $(N - 1)$ -dimensional Gaussian integrals with, generically, no closed form. The standard evaluator is the Geweke–Hajivassiliou–Keane (GHK) simulator (Geweke et al., 1994; Hajivassiliou and McFadden, 1998; Train, 2009), the econometric branch of the Genz separation-of-variables family (Genz, 1992). It treats the probability that alternative i beats the rest as an orthant probability of the difference utilities, estimated by sampling truncated normals along a Cholesky factor. It is elegant, stochastic, and resolved by draw count, and it prices one orthant per alternative, exploiting nothing of the product shared across the N winner events. Graduate students are warned off accordingly. Conlon’s industrial organization lecture notes put unrestricted MNP at “potentially J^2 parameters (that is a lot)!” and one simulated integral of dimension J per share evaluation: “I wouldn’t do this for $J \geq 5$.”³ The reference implementation of the logit tradition says the

²In racing parlance, we are recovering the relative ability of horses from the bookmaker odds, even when some horses are known to be correlated with others. The original transform (Cotton, 2021a) assumed independent performances.

³C. Conlon, graduate industrial organization lecture notes (multinomial choice), New York University,

same by omission. The word *probit* does not occur in the source tree of PyBLP (Conlon and Gortmaker, 2020), whose consumer heterogeneity is a rank- K_2 factor loading on observed characteristics, priced by its own documentation at 4^{K_2} product-rule quadrature nodes, which is 65,536 nodes at $K_2 = 8$.

The inversion is harder still, and the literature is quantitative about why. Berry, Linton and Pakes show that estimation through inverted shares requires simulation draws growing as the square of the number of products, because the inverse share map amplifies simulation and sampling error (Berry et al., 2004). Freyberger shows the surviving simulation error enters the estimator as a first-order bias (Freyberger, 2015). The standard escape at large N , estimation on a sampled subset of alternatives, rests on a logit cancellation, an explicit use of independence from irrelevant alternatives (IIA), and has no probit counterpart (Train, 2009). The dominant commercial conjoint platform, meanwhile, has shipped exactly this low-rank-plus-diagonal Gaussian race since 1999 (Huber et al., 1999), evaluating it by brute-force simulation with hand-tuned variance scalars, and matching target shares with an ad hoc fixed point its own authors call “fudge factors” and decline to encourage (Orme and Johnson, 2006).

The standard reason for needing none of this is the McFadden–Train theorem that mixed logit approximates any random-utility model arbitrarily closely (McFadden and Train, 2000). The construction is priced in its own paper. It sends the extreme-value scale to zero while the mixing distribution grows without bound, and the authors report “no practical indication of how to choose parsimonious mixing families, or how many terms are needed.” The Gaussian kernel they considered was set aside on computational grounds, not fidelity. The limit is real. The question is whether the required rank is affordable, and that is a computational question, not a theoretical one.

The field kept the computable models and gave up the model it started from. The concession is not free. One measure of the cost comes from an oracle factor-probit benchmark with every marginal variance held at one (Section 7). We generate a Gaussian factor race, calibrate both models to identical menu shares, delete an alternative, and ask where its demand goes. Among deletions of the largest alternatives, plain logit misallocates a median 25% of the released demand, against 1% for the correctly specified factor model (experiment 29).⁴ A companion factorial with a deliberately misspecified truth tells a similar story.

With unrestricted alternative intercepts, a single interior share vector cannot distinguish full-support random-utility models, because each can be inverted to match it. However, when logit, probit, or other models are calibrated to observed shares, the recovered utilities can serve as inputs to questions the data do not answer directly: elasticities, welfare, pricing, the effect of adding or removing an alternative, and combinatorial probabilities. It is at this point that they make different, falsifiable predictions.

This paper’s sole point is that probit’s computational barrier is not real for this covariance class. We solve the fixed factor-probit share calibration problem at $N = 1000$

github.com/chrisconlon/Grad-I0.

⁴Experiment numbers index committed, seeded scripts in the repository; see the Reproducibility section.

alternatives in three seconds at rank two with the compiled implementation, and ten thousand alternatives stay under a minute.⁵ The share map is also globally identified and onto the simplex interior (Proposition 2: shares determine utilities uniquely up to location).

Three ingredients make this possible:

1. *A fast forward map* (utilities $\rightarrow$ shares): all N probabilities at once in $O(QNL)$ for Q factor nodes and L lattice points. Calibration needs many forward evaluations; this is what makes them affordable.
2. *Resampling-free derivatives*: Jacobian–vector products in $O(QNL)$. The Jacobian is a weighted graph Laplacian, so reduced systems are symmetric positive definite. This supports implicit differentiation and Newton–Krylov solvers. The calibration reported below uses a cheaper damped Jacobi iteration.
3. *A single shared construction*: conditionally on the k factors (the rank) the alternatives are independent, and a lattice survival-product identity computes every alternative’s integral from one *shared survival field*, built once and reused for shares, slopes, and all N single-removal counterfactuals.

This one-pass identity comes from a mildly obscure but fitting literature: inferring ability from win probabilities in multi-entrant races (Cotton, 2021a). That work already emphasized very large fields and applications beyond the track, digital commerce among them. The central trick this paper leans heavily on was described there as a multiplicity calculus: a characterization of tie density on a lattice and a set of algebraic operations for combining races and removing a runner. That is perhaps why it has not been assembled for probit before, where the focus has been on more classical methods for numerical computation. No single ingredient here is new. The contribution is the combination: factor conditioning, all- N computation, derivatives, inversion, and implementation at $N = 10^4$.

2 The algorithm

We assume throughout a known factor structure. That structure is, as it happens, the one with which correlated utilities entered econometrics. Hausman and Wise (1978) induced correlation through random taste coefficients, giving a covariance that is exactly low-rank plus diagonal with observed attributes as loadings. Freeing the loadings is the factor-analytic MNP that McFadden (1984) proposed precisely to reduce the dimension of the integral.

⁵Estimating (V, D) is the outer problem. It is discussed in Section 8 but not claimed here.

Model. Throughout,

$$U_i = \mu_i + v_i^\top f + \sqrt{D_i} \varepsilon_i, \quad f \sim N(0, I_k), \quad \varepsilon_i \sim N(0, 1) \text{ i.i.d.}, \quad D_i > 0, \quad (1)$$

i.e. $\Sigma = VV^\top + \text{diag}(D)$. The chosen alternative is the argmax of U . The pair (V, D) is supplied and held fixed. Calibration solves for μ alone.

Conditional on the factors, the alternatives are independent, and the algorithm is built entirely on that fact. Write $g_i(\cdot | f)$ and $F_i(\cdot | f)$ for the conditional density and cumulative distribution function (CDF) of U_i given f . Alternative i wins when it beats every rival, so

$$p_i = \mathbb{E}_f \left[\int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx \right], \quad \sum_i p_i = 1, \quad (2)$$

the shares summing to one because ties are null for continuous laws. The N products need not be formed separately. Compute $\log F_{\text{field}} = \sum_j \log F_j$ once. Each product over $j \neq i$ is then recovered by subtracting the own term.

The forward pass proceeds as follows. Lay down one common grid of L points, wide enough to contain every runner’s distribution at every retained factor node. We call this interval the *envelope*. At each factor node f_q , shift each runner’s location by $v_i^\top f_q$ and evaluate each runner’s log-survival curve on the grid. This is the log-probability that runner i has not yet finished by position x . Sum these N curves to obtain the *shared survival field*, the log-probability that nobody has finished by x . To price runner i , subtract its own log-survival back out of the field, multiply by its density, and integrate along the grid. This is the probability that i finishes first at this factor node, with everyone else behind.⁶ Average over factor nodes with the quadrature weights, and normalize. Everything is done in the log domain so that deep-tail survivals never underflow. Algorithm 1 states this precisely.

A lattice of a few hundred points and a pruned product Gauss–Hermite rule suffice in practice. The exact resolution settings, and the evidence that they are converged, are collected with the numerical results in Section 4.

Algorithm 1 works in reflected *min-wins* coordinates $a = -\mu$, a race won by the smallest value, while the theory is stated in the natural *max-wins* coordinates μ ; the reflection keeps the loadings’ sign. It is valid because the Gaussian law is unchanged under $(f, \varepsilon) \mapsto (-f, -\varepsilon)$.

⁶This divide-one-out step is the “removing a runner” operation of the multiplicity calculus in Cotton (2021a), applied here once per alternative against a field built only once.

Algorithm 1 All-share forward pass (min-wins; reflected from argmax by $a = -\mu$)

Require: abilities a , loadings V , idiosyncratic variances D , factor nodes $\{f_q, w_q\}_{q=1}^Q$, lattice size L

- 1: $m_{iq} \leftarrow a_i + v_i^\top f_q$ for all i, q
 - 2: $[x_1, x_L] \leftarrow [\min_{i,q} m_{iq} - 8 \max_i \sqrt{D_i}, \max_{i,q} m_{iq} + 8 \max_i \sqrt{D_i}]$, uniform spacing Δx
 - 3: **for** $q = 1, \dots, Q$ **do**
 - 4: $\log S_i(x_\ell) \leftarrow \log \Phi(-(x_\ell - m_{iq})/\sqrt{D_i})$ for all i, ℓ $\triangleright$ never $1 - \Phi$
 - 5: $\log S_{\text{field}}(x_\ell) \leftarrow \sum_j \log S_j(x_\ell)$ $\triangleright$ one shared field
 - 6: $\tilde{p}_i \leftarrow w_q \sum_\ell g_i(x_\ell) \exp\{\log S_{\text{field}}(x_\ell) - \log S_i(x_\ell)\} \Delta x$ $\triangleright$ divide one out
 - 7: **end for**
 - 8: **return** $p = \tilde{p}/(\mathbf{1}^\top \tilde{p})$ and the diagnostic $|1 - \mathbf{1}^\top \tilde{p}|$
-

Per factor node, the conditional locations cost $O(Nk)$, the field log-product $O(NL)$, and each alternative's integral $O(L)$ after subtraction, so one complete forward pass is $O(QN(k + L))$, which is $O(QNL)$ for $k \ll L$. The same cached field also yields the full single-removal ensemble $\{P(j \text{ chosen} \mid i \text{ removed})\}_{i,j}$ in $O(QN^2L)$ arithmetic.

Inversion. Given target shares $\hat{p}$ and (V, D) , the inverse transform finds mean-zero μ with $p(\mu) = \hat{p}$.

Warm-start from a high-accuracy inversion of the *independent* race with matched total variances, which this same machinery provides using a single zero factor node. Then repeat the following. Run the forward pass at the current utilities. Read off each runner's own-slope $\partial \tilde{p}_i / \partial a_i$ from the same pass. Take a coordinatewise Newton step on log shares. Cap and damp the step. Re-center the utilities to mean zero. Algorithm 2 states the loop. Its convergence is analyzed in Section 3 (Proposition 4).

Algorithm 2 Calibration (damped Jacobi quasi-Newton)

Require: strictly positive target shares $\hat{p}$ (normalized), V , D , factor rule, tolerance 10^{-6} , cap 50 iterations

- 1: $a \leftarrow$ high-accuracy independent-race inversion using marginal variances $D_i + \|v_i\|^2$ (this same algorithm with a single zero factor node) $\triangleright$ warm start
 - 2: damp $\leftarrow 1$
 - 3: **repeat**
 - 4: run Algorithm 1 at a , collecting $\tilde{p}$ and the analytic own-location slopes $\partial_i = \partial \tilde{p}_i / \partial a_i$ from the same pass
 - 5: $r_i \leftarrow \log(\tilde{p}_i / \mathbf{1}^\top \tilde{p}) - \log \hat{p}_i$; residual $\leftarrow \max\{|r_i| : \hat{p}_i > 10^{-4}/N\}$
 - 6: halve the damping (floor 0.25) if the residual grew by more than 20%
 - 7: $a_i \leftarrow a_i - \text{damp} \cdot r_i / (\partial_i / \tilde{p}_i)$, steps capped at $\sqrt{D_i + \|v_i\|^2}$
 - 8: $a \leftarrow a - \bar{a}$ $\triangleright$ project to mean zero every update
 - 9: **until** residual $< 10^{-6}$
-

The own-location slopes are the diagonal terms that Proposition 3 later makes explicit.

The slopes in Algorithm 2 ignore the moving envelope and the output normalization.⁷

The tolerance is tail-aware. Alternatives with target shares below $10^{-4}/N$ are matched best-effort. By Proposition 2 they remain uniquely determined, but they are weakly resolved when target shares are estimated from finite data. Weak resolution is also a conditioning caution for the reduced Jacobian when shares are tiny or alternatives nearly collinear in their loadings.

Just as a horse at infinite odds sits outside the race, zero shares lie outside the proposition. The boundary targets correspond to utilities diverging to $-\infty$, so real assortment data require pseudocounts or regularization, not merely the implementation floor. This adjustment must be left to the designer of the application. For example, depending on the data at hand, extrapolation to approximate shares could utilize near-misses. Alternatively, the user can simply not lean on those inferred probabilities that are very small.

Proposition 4 establishes local linear convergence of the exact continuum iteration. We do not establish global convergence of the capped, adaptive-grid implementation. We can only say that in practice it typically converges in 10–15 iterations, and in 6–31 across the stress suite of Section 4. A globally safe fallback exists in principle. Calibration minimizes the strictly convex, coercive potential in the proof of Proposition 2, so a damped Newton–Krylov method with a line search converges globally for the continuum map by standard convex-optimization arguments. We ship the Jacobi iteration because it has never needed the safeguard.

Tail-stable implementation. Tail stability requires the log domain throughout. We evaluate $\log \Phi$ directly (forming $1 - \Phi(z)$ by literal subtraction rounds to zero near $z \approx 8.3$ through cancellation), use the analytic $\log g$, and form hazards as $\exp(\log g - \log \Phi)$. The Gaussian inverse Mills ratio⁸ is unbounded but grows only linearly in the extreme tail, rather than exponentially, which is the numerically important property.

A naive implementation that floors the density before taking its logarithm while the exact log-survival stays finite produces spurious e^{+92} hazards on problems as mild as $N = 8$ with heterogeneous D . That configuration is a regression test. In the forward map the same discipline lets deep-tail shares resolve rather than floor to zero.

Scope. The method needs the factor form. Where Σ has no adequate low-rank-plus-diagonal approximation in the choice-relevant quotient space (the part of Σ that moves choices, Section 7), simulation at exact Σ remains the tool (Section 7). The claim of scalability is in the number of alternatives at fixed modest rank. The quadrature size Q grows rapidly with the rank k (Section 4). At $k \leq 4$ the factor rule is deterministic Gauss–Hermite. Above that we use fixed-seed scrambled Sobol, so the pure quadrature story, like the speed, belongs to low rank. The advantages are precision, reproducibility, resampling-free derivatives, and the calibration machinery.

⁷They are Jacobi quasi-Newton preconditioners, not the exact diagonal of the adaptive map’s Jacobian.

⁸The inverse Mills ratio is $\phi(z)/\Phi(z)$, the ratio of the standard normal density to its cumulative distribution function. It is the conditional hazard in this setting, and behaves like $|z|$ as $z \rightarrow -\infty$ because $\Phi(z) \approx \phi(z)/|z|$ in the deep tail.

3 What the algorithm achieves

The algorithm computes a particular map from utilities to shares. This section records what that map is like: its identification and differentiability, the well-posedness of its inverse, the graph that organizes both, its derivative structure, and the convergence of the calibration loop. Interpretations connecting the map to older mathematics are collected in Section 5.

Proposition 1 (Identification and differentiability). *$p(\mu + c\mathbf{1}) = p(\mu)$, so $J = \partial p / \partial \mu$ has $\mathbf{1}$ in its null space and μ is identified only up to location. With B an $N \times (N - 1)$ basis of $\mathbf{1}^\perp$ and $\mu = B\eta$, share inversion and implicit differentiation are taken in reduced coordinates: $d\eta/d\theta = -(B^\top JB)^{-1} B^\top \partial p / \partial \theta$, provided $B^\top JB$ is nonsingular.*

Proof. For the invariance, write $U_i = \mu_i + \xi_i$ with $\xi_i = v_i^\top f + \sqrt{D_i} \varepsilon_i$. For every realization of the shocks, $\arg \max_i (\mu_i + c + \xi_i) = \arg \max_i (\mu_i + \xi_i)$, since adding c to every coordinate changes no pairwise comparison. The choice events therefore coincide realization by realization, so $p(\mu + c\mathbf{1}) = p(\mu)$ for all c . Differentiating this identity in c at $c = 0$ gives $J\mathbf{1} = 0$: the ones vector lies in the null space of J , and two utility vectors differing by a common shift produce identical shares, so μ is identified at most up to location.

For the reduced formula, suppose $\eta(\theta)$ satisfies the calibration identity $p(B\eta(\theta), \theta) = \hat{p}$ on an open set of θ . Total differentiation gives the N -equation system

$$JB \frac{d\eta}{d\theta} + \frac{\partial p}{\partial \theta} = 0.$$

Only $N - 1$ of these equations are independent, but none are lost by projecting: shares sum to one identically in both arguments, so differentiating $\mathbf{1}^\top p = 1$ gives $\mathbf{1}^\top J = 0$ and $\mathbf{1}^\top (\partial p / \partial \theta) = 0$; both terms already lie in $\mathbf{1}^\perp$, so the system is equivalent to its image under $B^\top$. That image is $(B^\top JB) d\eta/d\theta = -B^\top \partial p / \partial \theta$, and when $B^\top JB$ is nonsingular (established for this model in Proposition 2 below) it inverts to the displayed formula, with local existence and smoothness of $\theta \mapsto \eta(\theta)$ supplied by the implicit function theorem. $\square$

Proposition 2 (Shares determine utilities). *For model (1) with $D_i > 0$ and fixed known (V, D) , in max-wins coordinates, define for $i \neq j$*

$$w_{ij} = \mathbb{E}_f \int g_i(x | f) g_j(x | f) \prod_{\ell \neq i, j} F_\ell(x | f) dx > 0.$$

Then $J = \partial p / \partial \mu$ satisfies $J_{ij} = -w_{ij}$ and $J_{ii} = \sum_{j \neq i} w_{ij}$, i.e.

$$J = \sum_{i < j} w_{ij} (e_i - e_j)(e_i - e_j)^\top, \quad h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2.$$

This is a weighted graph Laplacian with everywhere-positive weights. Furthermore J is symmetric, positive semidefinite, has $\mathbf{1}$ as its only null direction, and $B^\top JB \succ 0$ for every full-rank basis B of $\mathbf{1}^\perp$. Moreover $p = \nabla G$ for the social-surplus function $G(\mu) =$

$\mathbb{E} \max_i \{\mu_i + \xi_i\}$. This is the Williams–Daly–Zachary/McFadden identity (Williams, 1977; Daly and Zachary, 1978; McFadden, 1981). The restriction of p to $\mathbf{1}^\perp$ is a smooth bijection onto the interior of the probability simplex. Every strictly positive target $\hat{p}$ has exactly one mean-zero utility vector.

Sketch. Off the diagonal μ_j enters p_i only through F_j , giving $J_{ij} = -w_{ij}$; translation invariance fixes the diagonal, so J is the stated weighted graph Laplacian, positive definite on $\mathbf{1}^\perp$ because every weight is positive. The inverse is the unique minimizer on $\mathbf{1}^\perp$ of the strictly convex, coercive potential $H(\mu) = G(\mu) - \hat{p}^\top \mu$, and the inverse-function theorem gives smoothness. Appendix A carries out each step. $\square$

Covariance is not fit. Surjectivity cuts both ways. For any fixed admissible (V, D) , every interior full-menu share vector is fitted exactly by some utilities, so a perfect same-menu fit carries no evidence that the covariance is right. The covariance is disciplined only by restrictions across menus, markets, or covariates, or by micro-level data (Chang et al., 2026). Its substantive content lies in the substitution and counterfactual predictions measured in Section 7.

The photo-finish graph. Given the origins of the central numerical insight, the reader will forgive us for speaking about Proposition 2 in racetrack terminology. Here w_{ij} is the probability density of a photo-finish between i and j , a tie for the win with everyone else behind, and moving utility from j to i moves share along every photo-finish edge at exactly that rate. The alternatives form a *photo-finish graph*: complete, with strictly positive weights, because Gaussian utilities give every pair a strictly positive tie density at the winning frontier.

The proposition’s conclusions are then graph facts. Constants span the null space because the quadratic form $\sum_{i < j} w_{ij} (h_i - h_j)^2$ sees only differences. Shares cannot identify the overall utility level, only gaps, which is economically obvious. The odds do not inform us of the class of the race.

Positive weights make the graph connected, so the constant vector is the *only* null direction, the gradient map is strictly monotone as an operator on $\mathbf{1}^\perp$ (not coordinatewise), and shares determine utilities uniquely.

And symmetry plus positive definiteness make every reduced Newton system of the continuum map symmetric positive definite, so conjugate gradients apply with the $O(QNL)$ Jacobian–vector product of Proposition 3 playing the role of a fast Laplacian multiply. Each linearization of the calibration solves a weighted graph-Laplacian system on the photo-finish graph, with weights that move with μ .

The graph is complete and its $O(N^2)$ weights are never materialized, so the benefit is matrix-free Krylov solving and spectral interpretation rather than the complexity guarantees of sparse-Laplacian solvers. The implemented map’s derivative differs from the continuum Laplacian by small normalization and grid terms, quantified under *the differentiated map* below.

At this point nothing is specifically Gaussian, just as nothing in Cotton (2021a) requires horse performance to be Gaussian. Any conditionally independent additive

random-utility race with everywhere-positive smooth densities has, under the corresponding integrability and differentiability conditions (finite social surplus, differentiation under the integral), a weighted-Laplacian Jacobian and an interior-simplex diffeomorphism modulo translation. Gaussian factor probit is the computationally important specialization.

Proposition 3 (Jacobian–vector products in $O(QNL)$). *With hazards $\lambda_j = g_j/F_j$, $\Lambda = \sum_j \lambda_j$, and $A = \sum_j h_j \lambda_j$ (all conditional on f , all functions of x), the Jacobian of the exact max-wins map acts on a direction h as*

$$(Jh)_i = \mathbb{E}_f \int g_i \left(\prod_{j \neq i} F_j \right) (h_i \Lambda - A) dx,$$

where the field sums Λ and A are formed once per node and lattice point, so all N components cost $O(NL)$ per node. In effect, a direction h perturbs alternative i 's share only through the gap between its own hazard weight $h_i \Lambda$ and the field average A . Translation invariance is visible ($h = \mathbf{1}$ gives $A = \Lambda$, hence $Jh = 0$), and the weighted-Laplacian form of Proposition 2 follows by expanding A . This enables Krylov solves of the reduced system in Proposition 1 without forming J densely at $O(QN^2L)$.

Proof. Write $R_i = \prod_{j \neq i} F_j$. For the own derivative, $\partial_{\mu_i} g_i(x) = -g'_i(x)$ (a location shift), and integrating by parts, $\int (-g'_i) R_i dx = \int g_i R'_i dx = \int g_i R_i \sum_{j \neq i} \lambda_j dx = \int g_i R_i (\Lambda - \lambda_i) dx$. For $j \neq i$, $\partial_{\mu_j} F_j = -g_j$, so the cross derivative is $-\int g_i g_j \prod_{\ell \neq i,j} F_\ell dx = -\int g_i R_i \lambda_j dx$. Summing against h , $(Jh)_i = \int g_i R_i [h_i (\Lambda - \lambda_i) - \sum_{j \neq i} h_j \lambda_j] dx = \int g_i R_i (h_i \Lambda - A) dx$; average over f . $\square$

The differentiated map. Because the algorithms presented here use lattice computation, we will be clear about the distinction between the approximation and the continuous idealization (just as in Cotton (2021a) we delineated the discrete horse race from the continuous).

Owing to differing conventions, three maps must be distinguished: the exact max-wins map $p^{\max}(\mu)$; the reflected min-wins map $p^{\min}(a)$ used by the implementation, related by $a = -\mu$ so that $D_\mu p^{\max}[h] = D_a p^{\min}[-h]$; and the implemented approximation $p_{Q,L} = \tilde{p}_{Q,L} / (\mathbf{1}^\top \tilde{p}_{Q,L})$, whose derivative carries a normalization correction: with $v = D\tilde{p}[h]$ and $s = \mathbf{1}^\top \tilde{p}$,

$$Dp_{Q,L}[h] = \frac{v - p_{Q,L} (\mathbf{1}^\top v)}{s}.$$

The correction vanishes for the continuum map and is numerically negligible at the resolutions tested. A further distinction is that integration by parts is a continuum identity. The Laplacian formula of Proposition 3 is therefore the exact derivative of the *continuum* map, not of the finite rectangle sum.

The exact derivative of the frozen-grid sum replaces the own term by $(x - m_i)/D_i$ (no integration by parts) and costs the same $O(QNL)$. That is why we have shipped it in the repository (`form="grid"`).

Measured at $L = 1501$ (the derivative-comparison resolution) the two forms agree to 1.7×10^{-14} . At $L = 51$ the continuum form is off by 2.6×10^{-5} while the grid form still

matches finite differences. At the production resolutions the difference is negligible. (In practice the Laplacian form is an accurate symmetric surrogate at the resolutions used.) But again, the exact differentiation of the normalized *frozen-grid* map is available when needed. The returned adaptive map additionally moves its envelope with μ , a tail effect measured below 10^{-8} at $L = 1501$.

The implemented inversion rebuilds the lattice envelope from the current utilities at each iteration, holds it fixed within the iteration, and projects the utilities to mean zero after every update (Algorithm 2). The same graph controls the convergence of that loop.

Proposition 4 (Local convergence of damped Jacobi). *For the exact continuum share map and a strictly positive target, the mean-zero damped Jacobi iteration underlying Algorithm 2 is locally linearly convergent for every fixed damping $0 < \alpha < 1$, and also at $\alpha = 1$ for $N \geq 3$, with asymptotic local convergence factor $\max_{j \geq 2} |1 - \alpha \lambda_j|$, the spectral rate on the mean-zero quotient, where λ_j are the nonzero eigenvalues of the random-walk normalized photo-finish Laplacian $\Delta^{-1}J$, $\Delta = \text{diag}(J_{11}, \dots, J_{NN})$.*

Proof. With $r(\mu) = \log p(\mu) - \log \hat{p}$, the Jacobian at the solution is $Dr = \text{diag}(p)^{-1}J$, and the iteration’s exact Jacobi preconditioner is $M = \text{diag}(J_{ii}/p_i)$, so the linearized update on the mean-zero quotient is $I - \alpha \Delta^{-1}J$. By Proposition 2, J is a graph Laplacian with strictly positive weights, so $\Delta^{-1}J$ is similar to the symmetric normalized Laplacian $\Delta^{-1/2}J\Delta^{-1/2}$, whose spectrum lies in $[0, 2]$ with 0 simple (the common-shift direction, removed by the quotient). The photo-finish graph is complete, hence nonbipartite for $N \geq 3$, so $\lambda_N < 2$ and $\max_{j \geq 2} |1 - \alpha \lambda_j| < 1$ for $0 < \alpha \leq 1$. Step caps are inactive near the solution. The projection changes the iterate only by a constant vector, so the iteration induces a well-defined linear map on the quotient $\mathbb{R}^N / \text{span}\{\mathbf{1}\}$, whose eigenvalues are the nonzero eigenvalues of $\Delta^{-1}J$. $\square$

Experiment 31 measures this at $N = 12$. The spectrum of $\Delta^{-1}J$ is $[0, 0.974, \dots, 1.379]$, and the observed contraction of the exact iteration matches the predicted factor to within 0.01 (0.369 against 0.379 at $\alpha = 1$; 0.316 against 0.318 at $\alpha = 0.7$). Global convergence of the capped, adaptive-grid implementation remains empirical; slow convergence corresponds to a small normalized spectral gap λ_2 , which the photo-finish graph makes interpretable. Tiny shares or nearly redundant alternatives weaken the graph’s connectivity.⁹

Beyond Gaussian factors. The factors need not be Gaussian because (2) requires only conditional independence and an integrable factor law. Other obvious candidates include Student, mixture, or empirical factor distributions. Also in keeping with Cotton (2021a) the base densities may be arbitrary and alternative-specific, modulo the numerical comments we make throughout that might apply to adversarially designed mixtures of Dirac distributions and the like intended to test the limits of the lattice computation.¹⁰

⁹In circuit terms, damped Jacobi preconditions each node by its own total conductance and ignores the wiring, so it is fast exactly when the network is well connected. Kirchhoff’s current law is the constraint that shares sum to one, and the irrelevance of a common voltage offset is the location non-identifiability.

¹⁰The implementation works in the reflected (min-wins) race, and $-U$ matches the reflected model only when the factor and base laws are symmetric. For asymmetric laws the reflected distributions must

4 Numerical evidence of accuracy

This section assesses the algorithm in its own terms, except where a comparator falls out of the same experiment. Systematic comparisons with other methods are deferred to Section 7. The evidence, in the order presented: correctness anchors on cases with closed forms; forward accuracy replicated across problem sizes (experiment 16); log-share accuracy against higher-resolution internal references (experiment 25); resolution sweeps of the two quadrature knobs (experiment 17); an error budget separating rigorous bounds from convergence evidence; calibration accuracy and its validation (experiments 16 and 21); an inversion validated against a methodologically independent RQMC reference (experiment 35); robustness and a grid stress test under extreme variance heterogeneity (experiment 33); compiled timings and scaling (experiment 19); derivative accuracy; and two by-products, estimation gradients (experiment 30) and the removal ensemble (experiment 18).

Settings. Unless otherwise stated, forward and calibration runs use $L = 501$ lattice points, chosen from the measured lattice sweep (experiment 17 spans $L = 101$ to 6001 , with self-convergence reaching machine precision by $L \approx 200$). The derivative comparison of Section 3 uses $L = 1501$, and higher-resolution reference calculations use $L = 3001$ to 8001 . The lattice sits on the adaptive interval $[\min_{i,q} m_i(f_q) - 8 \max_i \sqrt{D_i}, \max_{i,q} m_i(f_q) + 8 \max_i \sqrt{D_i}]$, where $q = 1, \dots, Q$ indexes the retained factor nodes. The rectangle rule uses Δx weights with no endpoint modification, differing from the trapezoidal rule only by negligible endpoint terms.

The main factor integration experiments use product Gauss–Hermite of order $15/11/9$ for $k = 2/3/4$. The boundary experiment (experiment 14) uses order 15 per dimension for all $k \leq 4$, which after pruning weights below 10^{-7} of the maximum leaves $Q = 145$, 1317 , and 10929 nodes for $k = 2, 3, 4$, making the exponential growth of Q with rank explicit. Scrambled-Sobol nodes (2^{13} points, fixed seed) are used only for $k > 4$. They are reproducible by construction, though unlike Gauss–Hermite the node set is not canonical. A different seed gives a different, equally valid rule. Survival floors sit at 10^{-300} and the quadrature output is normalized by its sum. The returned map is a reproducible, piecewise-smooth numerical approximation of (2). Statements about derivatives refer to derivatives of this approximation, with fidelity governed by Q and L .

Pruned factor weights are not renormalized explicitly. The final normalization conditions the quadrature on the retained node set, and the two contributions to the pre-normalization defect separate cleanly. The pruned factor weight is $\delta_{\text{factor}} = 2.39 \times 10^{-8}$ at $k = 2$ (analytic). The measured $|1 - \mathbf{1}^\top \tilde{p}| = 2.4 \times 10^{-8}$ equals it, so lattice truncation contributes below that level.

Benchmark designs. Unless a row states otherwise, the benchmark problems are drawn by one shared generator: $\mu_i \sim N(0, s^2)$, loadings i.i.d. $N(0, 0.25/k)$, and $D_i \sim U(0.5, 1.5)$, with seeds fixed in each script.

be used explicitly.

exp.	N	$k; s$	departures from the generator	targets / reference
16	20–1000	2; 1.0/1.5	$s = 1.0$ at $N \leq 200$, else 1.5	twin 2×10^6 -draw Monte Carlo (MC); 5×10^6 for inversion
17	200	2; 1.0	—	higher-resolution internal
19	1000–10000	2; 1.5	—	5×10^6 -draw MC
21	200, 1000	2; 1.5	robustness suite varies $s \in \{1, 1.5, 2\}$, loading scale $\{0.3, 0.7\}/\sqrt{k}$, $D \sim U(0.5, 1.5)$ or $U(0.1, 1.0)$, $k \in \{1, 2, 3\}$	lattice-generated and 5×10^6 -draw MC
25	$\leq 100, 1000$	2; 1.5	—	GH 60^2 , $L=5001$; 40^2 , $L=3001$
29	150	2; 1.0	rows normalized to $\ v_i\ ^2 = 0.36$; $D_i = 0.64$	5×10^6 -draw MC
33	20	2; 1.0	D log-spaced, ratios 10^2 and 10^3	$L = 8001$ and 2×10^7 MC
34	200	2; 1.0	loadings $N(0, 0.18)$; $D_i \sim U(0.6, 1.2)$	lattice at $L = 1501$
35	50–200	2; 1.0	—	independent RQMC at 2^{17}
36	50	2; 1.0	the misspecified factorial truth, twenty seeds	2×10^7 common draws per seed
38	200	2; 1.0	—	lattice at $L = 1501$

Correctness anchors. In the binary case, where MNP has a closed form, the lattice transform agrees to 3.3×10^{-16} (an easy anchor, useful for code validation rather than as evidence about the hard regime).

At $N = 5$ the lattice transform and a large-draw simulator both sit within the noise of a 10^7 -draw Monte Carlo reference, and the shipped package implementation agrees with the benchmark kernel to 1.1×10^{-5} (the package default uses its own coarser lattice and interval rule; at matched settings agreement is machine precision).

Error metric. We report two metrics. The first is the maximum absolute share error $\max_i |\hat{p}_i - p_i|$ against a simulation reference. Absolute error is not always the decision-relevant scale, though. In betting, moving a 100-to-1 chance to 150-to-1 is an enormous change carried by only $|\Delta p| \approx 0.003$. So we also report the log-share error $\max_i |\log p_i^{\text{lat}} - \log p_i^{\text{ref}}|$ over resolvable shares (which agrees with log-odds error to $O(p)$ on small shares). The log-domain lattice is at its strongest on this second metric. Deep-tail shares resolve to genuine small positives, whereas a direct simulation reference resolves relative error ϵ only for shares above roughly $1/(R\epsilon^2)$. A maximum over N coordinates tends to grow with N at fixed per-coordinate accuracy, and the reference’s own maximum-coordinate noise behaves likewise. Mean-absolute and total-variation summaries are reported alongside (experiment 16). At $N = 1000$ the lattice’s mean absolute error is 5.6×10^{-6} and total

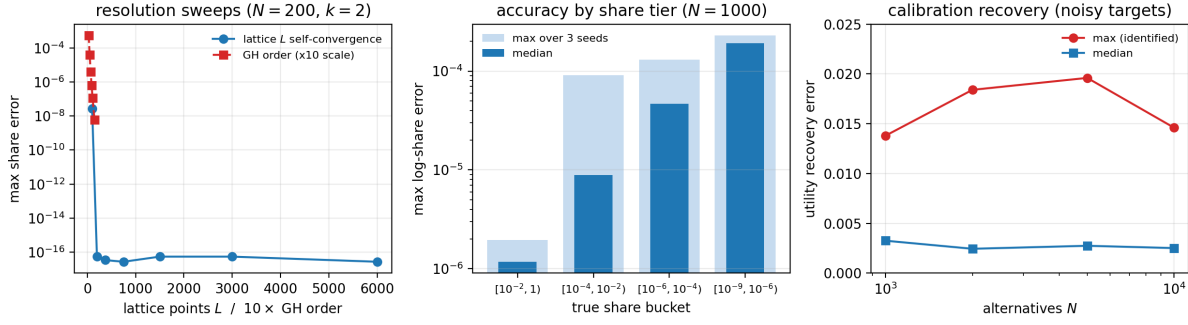


Figure 1: Central validation evidence, from the experiments. Left: lattice self-convergence in L and Gauss–Hermite factor sweep (experiment 17). Middle: log-share error against higher-resolution internal references by true-share bucket at $N = 1000$, three problems (experiment 25). Right: utility recovery against independent 5×10^6 -draw noisy targets over $N = 1000$ to 10000 (experiment 19); the plateau is target sampling noise, not solver error (experiment 21).

variation 2.8×10^{-3} against a fresh 2×10^6 -draw reference.

Replication. Ten problems per size at $N \leq 200$, three at $N = 1000$, each scored against twin independent 2×10^6 -draw references (experiment 16). The worst-case lattice error is $3.5\text{--}4.6 \times 10^{-4}$ at every size, at or below the references’ own replicate-to-replicate noise. The convergence evidence indicates accuracy far below this measurement floor (the resolution sweeps and the log-share assessment below).

Log-share assessment. Against higher-resolution internal references (Gauss–Hermite 60^2 with $L = 5001$ through $N = 100$; 40^2 with $L = 3001$ at $N = 1000$, three problems; experiment 25) the production settings’ log-share error is at most 8×10^{-5} on every share from 10^{-9} to one at $N \leq 100$. At $N = 1000$ it is at most 2×10^{-6} on shares above one percent, 9×10^{-5} down to 10^{-4} , and 2.3×10^{-4} on the deepest tails (shares to 10^{-9}). A plain winner-frequency simulation could not resolve these relative tail errors at practical draw counts: it needs about $1/(p\epsilon^2)$ draws for relative error ϵ on a share p .

Resolution sweeps. Figure 1 collects the central accuracy evidence. Experiment 17 sweeps each resolution knob separately at $N = 200$, $k = 2$. Holding the factor rule fixed, lattice self-convergence is 2.7×10^{-8} at $L = 101$ and at machine precision from $L \approx 200$. The convergence observed over the tested range is near-exponential. For these analytic, rapidly decaying integrands the lattice error becomes negligible very quickly, leaving the L -error subdominant.

Holding L fixed, the Gauss–Hermite sweep decays from 5.3×10^{-4} (order 3) to 5.9×10^{-9} (order 15).

Experimental errors. Empirical convergence evidence is not a rigorous proof, and we still rely on it in part. The error sources separate cleanly by status:

error source	status
factor-node pruning	rigorous bound (δ sup-norm)
finite envelope	rigorous bound ($N\Phi(-8)$)
covariance approximation	rigorous Kullback–Leibler (KL)/Pinsker bound (Section 7)
x -grid discretization	empirical convergence assessment
factor quadrature	empirical higher-order comparison
calibration residual	directly measured solver consistency

Stability checks and two genuine bounds. Self-convergence checks at large N show stability below what the simulation references can resolve: refining to ($L = 3001$, GH21) moves the shares by 1.2×10^{-8} at $N = 1000$ and 5.7×10^{-8} at $N = 5000$. This is a stability statement, not by itself an error bound, since both resolutions could share bias.

Two genuine bounds come free from the product structure. Conditional probabilities lie in $[0, 1]$, so pruning factor weight δ perturbs unnormalized shares by at most $\delta = 2.39 \times 10^{-8}$ in sup norm (measured 6.2×10^{-9}). The $\pm 8\sigma_{\max}$ envelope omits conditional mass at most $N\Phi(-8) + \Phi(-8)^N \leq (N+1)\Phi(-8) \approx 6 \times 10^{-12}$ even at $N = 10^4$ (conditional on each retained factor node, the lower tail by a union bound and the upper by conditional independence).

Calibration accuracy at $N = 1000$. At $N = 1000$, $k = 2$, with targets from 5×10^6 MC draws floored at 10^{-7} and renormalized, the inversion completes in about three seconds in the compiled implementation. The iteration solves its own equations to 6.4×10^{-9} . This is a statement about the solver, not statistical accuracy, since the target carries sampling noise near 2×10^{-4} . It recovers the utilities of the well-resolved alternatives (target share above 3×10^{-4}) to maximum error 0.017.

A replication study (experiment 16) repeats the inversion on a second $N = 1000$ problem against three independent 5×10^6 -draw targets: 161–162 alternatives well-resolved (98.1% of share mass), recovery error maximum 0.015–0.023 and median ≈ 0.0027 across replicates, with the recovery-versus-share curve reported alongside.

Validation: where the error comes from. Experiment 21 separates the error sources. A noiseless inverse test (targets generated by the lattice map itself, a deliberate inverse crime that isolates solver plumbing) recovers the utilities of every resolvable alternative to 6.5×10^{-7} at $N = 200$ and 1.2×10^{-7} at $N = 1000$ —solver error at the tolerance, so the ≈ 0.02 recovery in noisy tests is the targets’ sampling noise showing through. Alternatives with shares below the resolution floor are not chased, by design. Their tier is reported separately.

Independent inversion validation. Experiment 35 closes the gap between noisy simulation references and same-method lattice references. Target shares at $N = 50, 100$, and

200 are generated by the per-alternative factor-conditioned RQMC integral of Section 7 at 2^{17} scrambled-Sobol points, sharing no code path with the lattice, and the lattice calibration then inverts them. Utilities are identified up to location, and the tail coordinates carry the targets’ absolute-error floor of about 10^{-5} , so recovery is scored after recentering on the well-resolved set. On shares above 10^{-2} the recovered utilities match the truth to 7.6×10^{-5} , 1.1×10^{-4} , and 3.0×10^{-4} at the three sizes, with medians 1.3×10^{-5} to 6.8×10^{-5} . On shares above 10^{-3} the maximum is about 10^{-3} , which is the targets’ own relative accuracy at that share level. The recovery floor tracks the reference, not the solver.

Robustness. Twenty problems that vary utility spread, factor strength, D -heterogeneity, and rank all converged (6–31 iterations, recovery 0.004–0.024).

A further stress test targets heterogeneous idiosyncratic variances, where the common envelope is set by the largest $\sqrt{D_i}$ while the narrowest conditional density can approach the grid spacing. At variance ratio 10^2 the production lattice is already converged (log-share agreement 5×10^{-12} against $L = 8001$). At ratio 10^3 the production $L = 501$ carries a maximum log-share error of 1.0×10^{-3} , and $L = 2001$ restores machine precision (experiment 33). The practical rule is to scale L with the ratio of envelope width to the smallest conditional scale.

The assembled numerical Jacobian has the symmetry, row-sum, and reduced positive-definiteness properties predicted by Proposition 2. The reduced Jacobian assembled from N Jacobian–vector-product calls at $N = 50$ is symmetric to 1.7×10^{-18} (an assembly identity, tighter than the finite-difference symmetry check of Section 5), has row-sum defect 6.2×10^{-17} , and reduced eigenvalues in $[2.5 \times 10^{-5}, 0.22]$, positive definite as proved.

Compiled implementation. The same pass in Rust (in the repository, with machine-precision parity and identical iteration counts) calibrates $N = 1000$ in 3 seconds, $N = 5000$ in 18 seconds, and $N = 10000$ in 38 seconds (10, 11, and 13 iterations; independent 5×10^6 -draw targets, experiment 19).

Scaling. Calibration is approximately linear in N over the tested range (experiment 19), flat at 11–15 forward-pass equivalents of total wall time throughout (the excess over the iteration count is the warm start), with recovery quality unchanged (median error near 0.003, 96–99% of share mass identified). At fixed spatial resolution the cost grows slowly with the utility range, $O(\sqrt{\log N})$ under Gaussian utilities.

Figure 2 shows calibration wall time growing roughly linearly in the number of alternatives.

Derivatives. For fixed factor nodes and a fixed lattice, the transform is a reproducible, (piecewise-)differentiable function of μ whose derivatives are evaluated without resampling. The analytic own-location slopes of the unnormalized map come from the same pass and serve as the inversion preconditioner. Along a utility path at $N = 50$, the maximum second difference over dt^2 is 39.3 for fresh-draw GHK against 0.01 for both

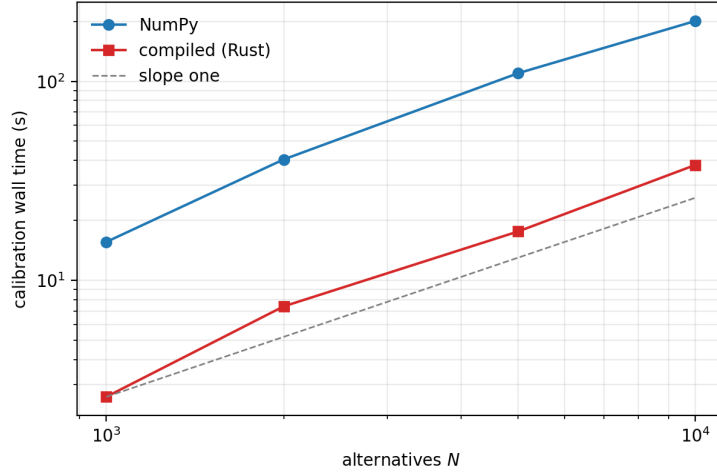


Figure 2: Wall time to calibrate the full utility vector from target shares ($k = 2$ factors, independent 5×10^6 -draw targets, experiment 19). Both implementations grow roughly linearly in N . The dashed guide has slope one. Recovery accuracy is assessed on alternatives above the share-resolution threshold.

GHK under common random numbers (CRN-GHK) and the lattice. The curves are in the committed output.

These are derivatives of the numerical approximation, not exact MNP derivatives, and GHK-based practice likewise differentiates its approximation, analytically or with fixed draws (Hajivassiliou et al., 1996; Hajivassiliou and McFadden, 1998; Train, 2009). The practical distinction is the character and controllability of the approximation error, quadrature resolution versus draw count, not the possibility of differentiation.

Estimation gradients (by-product). Experiment 30 parameterizes the covariance, $V(\theta) = \theta_1 V_0$ and $D(\theta) = e^{\theta_2} D_0$, across four markets that share one utility vector and differ by known factor-scale multipliers. The outer objective is the cross-market disagreement of the calibrated utilities, $\mathcal{L}(\theta) = \sum_m \|\mu_m(\theta) - \bar{\mu}(\theta)\|^2$, which vanishes at the truth. This is a cross-market restriction of the kind that disciplines the covariance where a single menu cannot. The gradient passes through the calibration fixed point by the implicit function theorem, $d\mu/d\theta = -B(B^\top JB)^{-1}B^\top \partial p/\partial \theta$. In this experiment the reduced Jacobian is assembled by symmetrized central differences and solved densely at $N = 50$. The matrix-free route through Proposition 3 is the large- N substitute. The implicit gradient matches central finite differences of the objective to 1.2×10^{-7} , and a damped Newton outer loop converges in three iterations to the truth up to the probit scale gauge (recovered $\theta_1^2 e^{-\theta_2} = 1.0000000$ against 1), which shares cannot identify. The scale ambiguity emerges from the experiment rather than being imposed.

Removal ensemble (by-product). The cached field also yields all N single-removal share vectors in $O(QN^2L)$, at roughly a third the wall time of recomputing them from

scratch at $N = 200$.

Against the strongest simulation comparator, which reuses each draw’s winner and runner-up to answer every removal at once in $O(RN + N^2)$, the deterministic ensemble is about $4.5\times$ more accurate at matched wall time at $N = 200$ (4.2×10^{-5} versus 1.9×10^{-4} ; experiment 18).¹¹

5 Further structure

This section records interpretations that connect the map of Section 3 to older mathematics. Nothing downstream depends on them.

Relation to the multiplicity calculus. The photo-finish weight is the differentiated form of the *multiplicity* that the original transform (Cotton, 2021a) tracks in the forward direction. There the multiplicity is a scalar field on the lattice: the expected number of runners sharing the winning cell, needed to split dead-heat payoffs, since an m -way tie pays each winner $1/m$.

Multiplicity so defined is the expectation of a discrete count. Its excess over one is $O(\Delta x)$ in the cell width and vanishes as the lattice refines. The weight w_{ij} is the continuum, pairwise density of the same event, a two-way tie between i and j with the rest behind, and it survives the continuum limit because $-w_{ij} = \partial p_i / \partial \mu_j$ is a rate rather than a count. The two are one identity apart: the multiplicity’s excess over one, per unit lattice spacing, is the total tie density

$$\sum_{i < j} w_{ij} = \frac{1}{2} \text{tr}(J),$$

which experiment 37 confirms—the measured excess-over-one divided by the cell width converges to $\frac{1}{2} \text{tr}(J)$ as $\Delta x \rightarrow 0$. The forward pass counts ties to divide payoffs. Calibration differentiates them into the graph-Laplacian weights that make it a network solve.

The conjugate view. Here we lean on social-surplus convex duality for discrete choice (Chiong et al., 2016; Norets and Takahashi, 2013). Our contribution is the algorithm, not the well-posedness of the inversion, which is known. G is convex with $p = \nabla G$, so $\Omega = G^*$ is a convex regularizer on the simplex and $p(\mu) = \arg \max_{q \in \Delta} \{\mu^\top q - \Omega(q)\}$. For i.i.d. Gumbel noise Ω is Shannon negentropy and the maximizer is softmax. In general it is the generalized entropy of Fosgerau et al. (2020), with the perturbation–regularization duality developed for machine learning by Berthet et al. (2020). Lin et al. (2026) estimate perturbed utility models by minimizing the associated Fenchel–Young loss, whose gradient in the utilities is exactly $p(\mu) - \hat{p}$; the forward pass here supplies that gradient for the factor probit case. Melo (2025) gives the online-learning reading of the same duality. Factor probit carries a covariance-dependent, non-separable entropy that couples alternatives through the factor structure.

¹¹This is a side capability, not the paper’s calibration emphasis.

In this paper we give three algorithms, and they could be thought of as three faces of that dual pair. The transform evaluates ∇G . Calibration evaluates the mirror map $\mu = \nabla \Omega(p)$ on the mean-zero quotient. The Jacobian–vector product applies the Hessian $\nabla^2 G$. Each is built from the same $O(QN(k + L))$ primitive. The forward map is one pass, each Jacobian–vector product is one pass, and each calibration iteration is one pass (total calibration cost multiplies by the iteration count, typically 10–15).

On the simplex, $\nabla \Omega(p)$ is an equivalence class modulo constants. Its mean-zero representative is the calibrated utility. The bijection itself is essentially known (Hotz and Miller, 1993; Norets and Takahashi, 2013; Berry et al., 2013), with the attributions detailed in Section 6. The connectivity mechanism of Berry et al. (2013) is the photo-finish graph’s. The appendix proof (Appendix A) keeps the Gaussian factor case self-contained and exhibits the weights the algorithm uses. Each identity in the proposition is also checked numerically in the repository: the w_{ij} representation against finite differences to 6×10^{-9} , symmetry to 4×10^{-12} , and positive definiteness of the reduced numerical Jacobian.

Transport form. There is another interpretation. On the contrast space, with $b_i = e_i - \mathbf{1}/N$ the vertices of a regular simplex, $\arg \max_i (x_i + \mu_i) = \arg \min_i \{\frac{1}{2}\|x - b_i\|^2 - \mu_i\}$: the choice cells are the Laguerre cells of quadratic semi-discrete optimal transport from the contrast Gaussian (the law of $P\xi$, $P = I - \mathbf{1}\mathbf{1}^\top/N$) to $\nu_q = \sum_i q_i \delta_{b_i}$, and Kantorovich duality identifies $\Omega(q)$, up to an additive constant, with $\frac{1}{2}W_2^2$ between them. Calibration is the dual solve of a structured transport problem. The weight w_{ij} is the Gaussian surface mass on the shared Laguerre facet (scaled by $1/\sqrt{2}$), and a Newton step is an electrical solve on that network, for which Proposition 3 is the matrix-free multiply.¹²

The geometry above is standard semi-discrete optimal transport. The power-diagram formulation goes back to Aurenhammer et al. (1998). Lévy (2015) optimizes the weights through a convex objective. Kitagawa et al. (2019) analyze a damped Newton method whose Hessian is a weighted graph Laplacian with facet-integral weights, with global convergence under regularity and support conditions that an unbounded Gaussian source does not immediately satisfy. Taşkesen et al. (2023) connect discrete-choice smoothing and semi-discrete transport directly. What is new here is the evaluation primitive. The factor structure gives an $O(QN(k + L))$ evaluation of all cell masses at once, and of Hessian–vector products without facet enumeration, when the source is a factor Gaussian and the sites are the simplex vertices.

At calibrated utilities the conjugate value admits a trace evaluation identity. The

¹²Place the alternatives as the corners of a regular simplex, and let each Gaussian draw pick its nearest corner, where corner i ’s reach is enlarged or shrunk by its utility μ_i . The choice regions are then the cells of a power diagram (a weighted Voronoi tessellation), and a choice probability is the Gaussian mass that falls in a cell. Calibration—nudging the utilities until each cell captures its target share q_i —is the semi-discrete optimal-transport problem of moving the Gaussian mass onto the corners at minimum squared-distance cost, with the utilities acting as transport prices and Ω as the transport cost. The weight w_{ij} is the Gaussian mass on the wall between cells i and j ; a Newton step pours the share residuals into the nodes as currents, treats those wall-masses as conductances, and solves for utility corrections as voltages, which Proposition 3 performs without forming the matrix.

identity is a factor-probit specialization of Gaussian integration by parts, Stein’s identity (Stein, 1981). Its useful content here is the photo-finish decomposition and the fact that the same matrix-free primitives the calibration uses evaluate it.

Proposition 5 (Conjugate trace identity). *For model (1), with $J = J(\mu)$ and $p = p(\mu)$,*

$$G(\mu) = \mu^\top p + \text{tr}(\Sigma J), \quad \Omega(p) = -\text{tr}(\Sigma J) = -\sum_{i < j} w_{ij} s_{ij}^2, \quad s_{ij}^2 = D_i + D_j + \|v_i - v_j\|^2.$$

Sketch. $G - \mu^\top p = \mathbb{E}[\xi_I]$, the mean shock of the winner. The divergence theorem turns each choice cell’s contribution into Gaussian facet integrals, and the facet masses are the photo-finish weights w_{ij} . Summing gives $\text{tr}(\Sigma J)$. Appendix B carries out the steps. $\square$

Read the graph as a network of wires: w_{ij} is the conductance of edge ij , and $s_{ij}^2 = \text{Var}(U_i - U_j)$ is the noise in the i -versus- j race. The entropy Ω is minus the total conductance, each edge weighted by its noise. Only differences $v_i - v_j$ and sums $D_i + D_j$ appear, so adding a common part to the covariance leaves Ω unchanged. Only the contrast part $P\Sigma P$ of the covariance can matter, a property Section 7 develops for covariance fitting.

We checked this in the code (see experiment 27 in particular). The binary closed form $\Omega = -s \phi(\Phi^{-1}(q))$, with s^2 the difference variance and q the first share, is verified to 3×10^{-13} . The full identity is verified to 3×10^{-9} . That G^* is an assignment problem was already known (Chiong et al., 2016); new here is the simplex form and also the evaluation of the conjugate value at calibrated utilities from k Jacobian–vector products and the diagonal slopes. The gradient $\nabla\Omega(p) = \mu$ is dearer. It requires the calibration solve, and Hessian actions of Ω require reduced linear solves built from the same Jacobian–vector products.

6 Related work

Prior art. The nearest methods, and what each lacks:

approach	all N shares	calibration
GHK simulation (Geweke et al., 1994; Hajivassiliou and McFadden, 1998)	N separate runs, $R^{-1/2}$ (plain MC)	no reported large- N share inversion known to us
direct utility simulation	whole vector per draw, $R^{-1/2}$	raw frequency map is nonsmooth
analytic approximations (Bhat, 2011; Connors et al., 2014)	per-alternative	not provided
factor quadrature (Butler and Moffitt, 1982; Elrod and Keane, 1995)	N separate $(k+1)$ -dim integrals	not provided
Bayesian factor MNP (Loaiza-Maya and Nibbering, 2022)	simulation inside estimation	posterior, not a share map
neural amortization (Huch and Keane, 2026)	learned approximation, accuracy not resolution-controlled	not provided
BLP-side inversion (Berry et al., 1995; Conlon and Gortmaker, 2020)	—	logit family only
this paper	one $O(QNL)$ pass	Prop. 2; 3 s at $N=1000$ (compiled)

GHK and simulated likelihood are standard in econometrics (Geweke et al., 1994; Hajivassiliou and McFadden, 1998; Train, 2009), with Hajivassiliou et al. (1996) the standard treatment of simulated multivariate normal (MVN) probabilities and their derivatives; Genz (1992) developed the same separation-of-variables construction in numerical statistics, where its RQMC form is the default MVN evaluator, anchored by the Genz–Bretz book (Genz and Bretz, 2009) and the `mvtnorm` package (Mi et al., 2009).

Deterministic grid recursions exist within that ecosystem: Miwa et al. (2003) evaluate orthant probabilities by recursion on one-dimensional lattices, at factorial cost in the dimension for general correlation, and Craig (2008) accelerates the Markov case; both price one probability per run. Ridgway (2016) reads GHK as sequential importance sampling, proves its variance can deteriorate exponentially, and builds sequential Monte Carlo (SMC) alternatives. Botev (2017) sharpened the family with minimax tilting, and Genton et al. (2018) began the hierarchical line in this journal. Cao et al. (2021) push MVN probabilities to thousands of dimensions with tile-low-rank QMC, machinery pointed almost exactly at our difference covariances, which are diagonal plus rank $k + 1$. Cao and Katzfuss (2026) reach linear per-probability cost past twenty thousand dimensions with Vecchia approximations.

Deterministic approximations run from Mendell–Elston (Mendell and Elston, 1974) (surprisingly competitive when optimally ordered (Connors et al., 2014)) through Bhat’s MACML (Bhat, 2011) to expectation propagation: Cunningham et al. (2011) found EP highly accurate precisely for rectangular regions, and Fasano and Denti (2025) evaluate generic Gaussian CDFs by EP on a dual probit model, deterministically and sampling-free.

Every one of these prices one orthant at a time. None couples the N winner probabilities.

Factor dimension reduction for probit is old, and older than econometrics: Curnow and Dunnett (1962) showed that product correlation collapses the multivariate normal integral to a one-dimensional integral of products of univariate CDFs, the reduction this paper conditions on, and Dunnett (1989) is its standard algorithm. The ranking-and-selection tradition built on the same integrals from the 1950s (Bechhofer, 1954; Gupta, 1963), always evaluating one designated candidate per integral: assembling all N that way costs N separate integrals where the shared field gives all N in one pass.

In econometrics, Butler and Moffitt (1982) integrated one factor by quadrature, and Lee (2000) stabilized that quadrature numerically; McFadden (1984) proposed factor-analytic MNP expressly as dimension reduction; Stern (1992) used the same conditional-independence product of normal CDFs with Monte Carlo over the residual; Bolduc (1999) handled large choice sets with parsimonious covariances but simulated probabilities. Sparse-grid integration (Heiss and Winschel, 2008) attacks the rank growth that binds product rules.

Elrod and Keane (1995) estimated factor-analytic probit for panel data; Haaijer et al. (1998) estimated utility covariances in conjoint MNP; Cohen (2010) built a structured covariance probit expressly for demand estimation; Loaiza-Maya and Nibbering (2022) pursue large choice sets by Bayesian estimation, with variational follow-ups (Loaiza-Maya and Nibbering, 2023); Kim et al. (2026) reach twenty alternatives at a million observations by neural variational inference, and Ding et al. (2024) exceed a hundred with expectation-propagation E-steps. All are approximate forward maps for estimation, and none inverts shares. Girolami and Rogers (2006) use the one-dimensional product identity in its diagonal-covariance form, per class. Collier et al. (2021) place the factor-Gaussian race with low-rank-plus-diagonal covariance on the last layer of large-scale image classifiers, including thousand-class ImageNet, and approximate the winner probabilities by Monte Carlo over a temperature-softmax relaxation; Berthet et al. (2020) differentiate general perturbed-argmax layers by Monte Carlo, with $p = \nabla G$ as the organizing identity. Both simulate the map this paper evaluates deterministically.

Racing supplied two of our ingredients. Henery (1981) wrote the survival-product win integral for normal running times and gave its first-order analytic inversion. Ali (1998) inverted market odds to abilities numerically at eight-runner fields, and betting practice otherwise approximated the inversion away (Lo and Bacon-Shone, 1994). The lattice survival-product identity and its fast inverse transform come from racing (Cotton, 2021a) and ship in the `winning` package (Cotton, 2021b). Cotton (2021a) names the common-factor extension developed here as an open problem.

The shared-product assembly itself also appears in competing-risks computation: Mahani and Sharabiani (2019) compute all cause-specific cumulative incidences on one shared grid by forming the product of every survival function once and dividing out each cause’s own factor (software since 2015), and the Aalen–Johansen estimator (Aalen and Johansen, 1978) reads all causes off a single survival product nonparametrically. Neither embeds the assembly in factor quadrature, differentiates it, nor inverts it. Racing arrived at it independently.

An older transportation literature used “calibration” for MNP parameter estimation.

Daganzo et al. (1977) introduced an early numerical MNP forecasting algorithm and claimed applicability to large problems, with Daganzo (1979) the monograph treatment; Sheffi et al. (1982) reviewed probability and calibration methods and proposed simulation-based estimation; Kamakura (1989) embedded the Mendell–Elston approximation in a calibration routine. These estimate parameters from choice data or approximate designated probabilities; none provides a smooth all- N share map, a shared-field Jacobian–vector product, or aggregate-share inversion at the scale considered here.

Inversion theory is likewise established. Berry (1994) made share-to-utility inversion the workhorse of demand estimation; Bonnet et al. (2022) invert demand in general, including nonsmooth random-utility models, through two-sided matching; Hotz and Miller (1993) inverted choice probabilities to utility differences; Norets and Takahashi (2013) prove surjectivity of the utility-to-probability map; Berry et al. (2013) obtain demand invertibility from connected substitution; Chiong et al. (2016) invert through the convex conjugate; Fosgerau et al. (2020) show the probit inverse exists by generalized-entropy duality, without an algorithm.

Berry et al. (1995) built demand estimation on the logit-side inversion, and Conlon and Gortmaker (2020) maintain its modern practice. Chia (2021) differentiates logit BLP automatically. On the generalized extreme value (GEV) side, closed-form flexible-correlation kernels pursue rich substitution without high-dimensional integration (Tinessa et al., 2020; Tinessa, 2021). Meanwhile Huch and Keane (2026) amortize correlated-choice probabilities with neural networks, evidence of current demand for this computation.

The novelty here is accordingly not dimension reduction (Curnow and Dunnett, 1962), nor the order-statistic integral, nor the shared-product assembly in isolation (Mahani and Sharabiani, 2019; Cotton, 2021a), nor well-posedness of inversion. It is the all-alternatives assembly in $O(QNL)$, its large- N implementation, the $O(QNL)$ Jacobian–vector product, and the shared-field reuse for calibration and counterfactuals. Convenient error partitioning (Train, 1995) integrates part of the noise away for one designated alternative at a time. The shared field recognizes that the N resulting integrands share one product and computes the whole vector together.

7 Comparisons with existing algorithms

We now compare with existing approaches: first the natural factor-aware competitor, then simulation in its weakest and strongest forms, then the boundary regimes where the factor form itself, rather than the numerics, is the binding approximation, and finally the substitution experiments that measure what logit’s restriction costs.

The natural competitor. Conditioning on the factors and the alternative’s own shock makes each probability a $(k+1)$ -dimensional integral,

$$p_i = \mathbb{E}_{f,z} \prod_{j \neq i} \Phi \left(\frac{\mu_i - \mu_j + (v_i - v_j)^\top f + \sqrt{D_i} z}{\sqrt{D_j}} \right),$$

evaluable per alternative by modern randomized quasi-Monte Carlo: the natural competitor that uses exactly the structure this paper assumes. The conditioning idea is the *convenient error partitioning* of Train (1995), which conditions on components common across alternatives and computes what remains analytically. It still prices N separate integrals, $O(RN^2)$ in total, where the shared field couples them into one $O(QN(k + L))$ pass. Measured at matched 4×10^{-4} accuracy (experiment 24; the two implementations first verified against each other to 1.7×10^{-5}), the per-alternative route is about $140\times$ slower than the single shared-field pass at $N = 1000$, a gap that widens linearly in N . Both implementations are NumPy on the same machine.

The price of smoothness. Smoothness was never a side issue in this literature. It shaped it. The raw frequency simulator is a step function of the parameters, so Stern (1992) accepted a large per-draw cost purely to smooth it, and GHK’s adoption owed as much to producing smooth, positive, differentiable estimates (Hajivassiliou and McFadden, 1998) as to its variance reduction. The lattice ends that trade. It is smooth to resolution, and its derivatives are exact integrals of the same pass, so the property the simulation literature paid variance and wall time for costs nothing here.

The limits of winner-frequency simulation. With common random numbers a smoothed simulator such as CRN-GHK is a fixed function that a solver can match to fine tolerance (a raw draw-and-argmax map is piecewise constant on the $1/R$ grid and cannot be). The limitation is not smoothness. It is the error of that fixed function against the exact map, and the cost of driving it down.

Its coordinatewise error against the exact map is of order $\sqrt{p_{\max}/R}$, so making the simulated map itself τ -accurate costs $R \approx p_{\max}/\tau^2$ independent draws per evaluation, and inverting it exactly still leaves utilities whose true shares miss the target at a scale that is order τ but amplified through J^{-1} .

If we take the example of crude winner-frequency simulation with $N = 5000$, this means driving the simulator to the solver’s consistency level, $\tau \sim 10^{-6}$, well beyond the 2×10^{-4} statistical accuracy of the noisy-target experiments. At that scale each forward evaluation needs $R \approx p_{\max}/\tau^2$ draws, which at the direct-simulation throughputs we measured is weeks of computation. A calibration could take months to a year.

If calibration time at a fixed accuracy can be reduced from months to a few seconds, the tradeoff between probit and its analytically simpler logit cousin changes. This particular example is roughly a six-order-of-magnitude speedup.

Common random numbers do not change the plain-MC rate. Randomized QMC and specialized Gaussian integration can improve rates as well as constants. The structural gap is different: per-alternative formulations price N separate integrals whose integrands each carry $O(N)$ factors, and the shared field removes that factor of N . A maximum over N coordinates adds a further $\log N$ factor in high-probability statements.

Returning to winner-frequency calibration, at the crude $\tau = 10^{-4}$ where simulation-in-the-loop first becomes feasible, the calibrated model’s true shares miss the target at 10^{-4} and the answer moves with each set of samples.

In contrast the quadrature route’s accuracy against the exact map is set by resolution, not draws, at cost independent of τ until resolution binds. We are unaware of a reported probit share inversion at this scale in simulation-based practice, and this scaling is presumably why. Within the fixed low-factor regime, that constraint is now gone.

The strongest simple simulator. Simulation can do much better than winner frequencies at no structural cost, again following Train (1995). Draw the factors and every idiosyncratic shock, form each alternative’s best rival $M_{-i} = \max_{j \neq i} U_j$ (all N of them from the top-two statistics, in $O(N)$ per draw), and average $\Phi((\mu_i + v_i^\top f - M_{-i})/\sqrt{D_i})$. The estimator is unbiased coordinatewise, smooth apart from max-switch kinks, and $O(RN)$ for the whole vector. Measured at $N = 200$ (experiment 38), at $R = 10^5$ it beats winner frequencies by roughly $3\times$ in maximum absolute error and $6\times$ in total variation (3.2×10^{-4} and 1.4×10^{-3}). Its best log-share error on shares above 10^{-3} is 1.8×10^{-2} , against about 10^{-4} for the lattice at comparable sizes. Conditioning improves the absolute scale but not the relative accuracy of small shares, and calibration needs the latter.

A smooth all- N comparator from machine learning is the temperature-softmax relaxation $\mathbb{E}[\text{softmax}((\mu + \xi)/T)]$ with common draws (Collier et al., 2021; Berthet et al., 2020), computing every share in $O(RN)$. It is biased at any fixed temperature and noisy at finite draws. Measured at $N = 200$ against the lattice map (experiment 34), its best settings over $T \in [0.03, 1]$ and $R \leq 2 \times 10^5$ leave a maximum absolute share error above 9×10^{-4} and log-share errors above 0.1 on shares over 10^{-3} , with total variation 0.43 at $T = 1$. Sharpening T trades bias for variance. The lattice’s corresponding log-share error is below about 10^{-4} at the sizes bracketing $N = 200$ measured in experiment 25.

General covariance through the factor approximation. To use the method when Σ is general, one first fits a factor model. Absolute utility levels are arbitrary, just as only relative horse ability matters in a race, and failing to account for this can “waste” factors.

To be more formal, the method requires $\Sigma \approx VV^\top + \text{diag}(D)$. GHK does not. Choices depend on (μ, Σ) only through $P\mu$ and $P\Sigma P$, $P = I - \mathbf{1}\mathbf{1}^\top/N$, because the argmax is determined by pairwise differences. So a common factor loading $V \rightarrow V + \mathbf{1}c^\top$ shifts every conditional location equally and cannot move the argmax (a machine-precision identity).

Our advice is that a factor fit must be run in a choice-relevant quotient space. In contrast, fitting raw Σ can spend scarce rank on an irrelevant common component ($\Sigma = \kappa^2 \mathbf{1}\mathbf{1}^\top + \beta\beta^\top + \text{diag}(D)$ with large κ : the raw rank-1 fit takes $\mathbf{1}\mathbf{1}^\top$ and misses $\beta\beta^\top$ entirely; on such an example the contrast-space fit cuts rank-1 share error from 4.6×10^{-2} to 8.3×10^{-3}).

The asymmetry matters. Only the common *factor* direction is irrelevant. A common addition to D inflates every difference variance and is choice-relevant, so D must come from the quotient fit.

The suggested fit is an iterative principal-factor heuristic applied to $P\Sigma P$ (eigendecompose $P\Sigma P - \text{diag}(D)$, take the top k directions, refit D from the diagonal with a 10^{-3} floor, iterate to 10^{-10}), followed by centering and singular value decomposition (SVD) canonicalization of the loadings (ordered columns, fixed signs).

This procedure will not, however, minimize a projected norm ($P \text{diag}(D) P$ is not diagonal). Therefore the repository also implements an alternating least-squares fit with exact block updates on the reduced quotient (precisely, a top- k positive semidefinite (PSD) step for the loadings and a nonnegative least squares for D that is exact block minimization but should not be interpreted as global optimality). On the boundary matrices it changes the quotient residual by under 1%.

The covariance-quotient residual $\|P(\Sigma - \hat{\Sigma})P\|_F / \|P\Sigma P\|_F$ is reported alongside every share error in the committed output.

We test this approach on correlation matrices with spectral decay $\lambda_m \propto m^{-\gamma}$, $\gamma \in \{0.5, 1.5, 3\}$, at $N = 50$ (sharing one orthogonal eigenbasis across γ so decay comparisons are not confounded with basis realization). The power law holds before diagonal standardization, which perturbs the spectrum; the actual leading eigenvalues (3.7, 17.1, 34.2 at $\gamma = 0.5, 1.5, 3$) are printed by the script.

Share error is measured against an 8×10^6 -draw reference, with GHK at the exact Σ for comparison. The plotted quantity is the rank- k approximation error of the stated procedure, not a minimum over factor models.

Rank-8 errors: 1.8×10^{-3} , 3.7×10^{-3} , and 3.1×10^{-3} at $\gamma = 0.5, 1.5, 3$ from $k = 1$ starting points of 2.3×10^{-3} , 1.5×10^{-2} , 7.1×10^{-2} .

The lattice versus simulation discrepancy is also tested. We simulate under the *fitted* factor model $\hat{\Sigma}$ and compare with the lattice at the same $\hat{\Sigma}$. That discrepancy, which includes the finite reference's own noise, is flat at $2\text{--}7 \times 10^{-4}$ across every (γ, k) tested. The remaining error against the exact- Σ reference is therefore mostly covariance-fit error. The quadrature is not the binding constraint.

Against GHK at $R = 10^4$ (3.2, 2.3, 4.3×10^{-3}), rank-8 wins at $\gamma = 0.5$ and 3 but *loses at intermediate decay* $\gamma = 1.5$ (3.7×10^{-3} versus 2.3×10^{-3}). This is the hardest example. Our interpretation is that the diagonal standardization perturbs the nominal power law. The leading eigenvalue varies from 3.7 to 34.2 across the three cases.

We also tried replication over three independent eigenbases. Rank-8 beats GHK at $R = 10^4$ in 8 of 9 basis-decay cases, with the single loss at intermediate decay. The intermediate construction seems to be getting close to the boundary where GHK should be preferred, but it is not clear cut. The wall time is not matched. Our Sobol-node rank-8 pass takes about 20 seconds versus about one second for GHK, so the latter seems to be the cheaper route to middling accuracy (the lattice's advantage remains at large N and small k). So where accuracy demands exceed the achievable rank- k error, GHK's arbitrary- Σ generality means that (forgive the pun) it should still be the tool of choice.

A forward-share certificate for the covariance fit. How much can a low-rank covariance approximation move the shares? Only the contrast part of the covariance matters, so replacing Σ by a fitted $\hat{\Sigma}$ changes any share by at most

$$|p_i(\Sigma) - p_i(\hat{\Sigma})| \leq \sqrt{\frac{1}{2} \text{KL}(\mathcal{N}(B^\top \mu, B^\top \Sigma B) \parallel \mathcal{N}(B^\top \mu, B^\top \hat{\Sigma} B))}.$$

This is Pinsker's inequality, with B any basis of $\mathbf{1}^\perp$. The Gaussian KL has a closed form. This turns the covariance-fit error into a guaranteed bound on the share error, computed

before the transform is ever run. The bound controls the forward map at fixed μ . It does not bound recalibrated utilities or deletion predictions, which pass through the inverse map. The bound is loose, about two orders of magnitude (experiment 26). Fitting two factors to a fast-decaying four-factor truth guarantees a share error below 5.3×10^{-2} when the true error is 4×10^{-4} , and three factors guarantee 8.4×10^{-3} against a true 9.7×10^{-5} .

Substitution: where a deleted alternative’s demand goes. When an alternative is removed, logit must spread its demand over the survivors in proportion to their existing shares—the independence of irrelevant alternatives (IIA) property—while a factor model can send it preferentially to close substitutes. These experiments measure which is right under known synthetic truths, and by how much.

The primary comparison, experiment 29, is an oracle structural benchmark. For each of twenty seeds the truth is a Gaussian factor race at $N = 150$, $k = 2$: utilities $U_i = \mu_i + v_i^\top f + \sqrt{0.64} \varepsilon_i$ with $\mu_i \sim N(0, 1)$, standard Gaussian f and ε_i , and loading rows drawn Gaussian then normalized to $\|v_i\|^2 = 0.36$, so every marginal variance is exactly one and the design moves only the off-diagonal covariance. The factor-probit candidate is correctly specified. It is calibrated to the observed menu shares using the true (V, D) , while plain logit reallocates proportionally from the same shares.

The experiment therefore isolates the pure substitution cost of imposing independence of irrelevant alternatives under oracle loadings. It is not an experiment in loading estimation or in robustness to misspecification. The factorial below covers misspecification. Targets are independent 5×10^6 -draw simulations, so there is no inverse crime. Median misallocation of released demand (logit against factor probit) is 0.245 against 0.009 for deleted shares above 10%, 0.275 against 0.031 at 2–10%, and 0.325 against 0.113 at 0.5–2%, with factor probit winning every paired seed in each of those strata (nineteen seeds produce a deletion above 10%; twenty elsewhere). Below 0.5% both degrade (0.78 against 0.68, 17 of 20) as the targets’ resolution binds. These are the numbers quoted in the introduction.

A richer factorial design varies the noise family as well, at the cost of unequal marginal variances. We vary two axes in a 2×2 design, the factor structure and the idiosyncratic noise family, holding the idiosyncratic scale common so the two axes stay clean (coupling either with alternative-specific scales would confound it: an alternative-specific-scale Gumbel race is not mixed logit, and heteroskedasticity alone already breaks IIA).

Ground truth ($N = 50$) is misspecified for every candidate: $t(5)$ factors (unit variance) and skew-normal idiosyncratic noise (shape $a = 3$), standardized to mean zero and unit variance, homoskedastic, with $\mu^* \sim N(0, 1)$ and oracle loadings $v_{ir}^* \sim N(0, 0.36/k)$, $k = 2$. (The race is computed min-wins, so positive skew in race coordinates is *left*-skew in utility coordinates.)

The four candidates hold the idiosyncratic scale fixed and vary $\{\text{Gumbel, Gaussian}\}$ base $\times \{V = 0, V = V^*\}$. The Gumbel base is variance-standardized to one, so the family axis does not alter the factor-to-idiosyncratic variance ratio.

Because the implementation is min-wins, the Gumbel base is the *reflected* (minimum) Gumbel corresponding to a standard maximum-Gumbel in utility coordinates. The code

verifies that its zero-loading case equals softmax at $N = 12$ to 2.8×10^{-17} . The candidate factor law is Gaussian in all cases.

This is an *oracle-loading* design: both factor candidates receive the true V^* rather than estimating it. All four are calibrated to the same full-menu shares (residuals $\leq 4 \times 10^{-10}$).

Deletion truths come from 2×10^7 *common* utility draws with the top three alternatives retained per draw, from which every single-deletion and pair-deletion truth follows with common random numbers.

Of 150 deletion blocks (50 singles, 100 fixed-seed random pairs), 45 with deleted mass below 0.05% are excluded as unresolvable; of the 105 scored blocks, 83 fall in the three best-resolved strata (26/29/28 at $>10\%$ / $2\text{--}10\%$ / $0.5\text{--}2\%$) and the remaining 22, with deleted mass between 0.05% and 0.5%, appear in the final column (their truths sit closest to the common-draw resolution floor, so that column carries the widest error bars).

The score per deletion block $\mathcal{R}$ is the misallocated fraction of the redistributed mass,

$$\text{TV}\left(q_{\text{model}}^{(-\mathcal{R})}, q_{\text{truth}}^{(-\mathcal{R})}\right) / \sum_{i \in \mathcal{R}} p_i,$$

averaged within strata:

	mass $> 10\%$	2–10%	0.5–2%	0.05–0.5%
independent Luce (= plain logit)	0.229	0.245	0.261	0.305
independent probit	0.210	0.232	0.230	0.267
factor mixed logit	0.114	0.146	0.199	0.244
factor probit	0.045	0.062	0.094	0.116

The factorial decomposition uses the two best-resolved strata, reported as a pair in each set of parentheses. The factor increment is dominant in both families ($+0.115/+0.099$ within Gumbel, $+0.165/+0.170$ within Gaussian). The family increment alone is small ($+0.019/+0.013$ at $V = 0$) but grows with factors present ($+0.069/+0.084$). The interaction is negative ($-0.050/-0.071$), so factor structure and the Gaussian base are complements in this design.

One caveat applies to the factorial. Its loadings $v_{ir} \sim N(0, 0.36/k)$ add $\|v_i\|^2$ to each marginal variance, unevenly across alternatives, so its factor axis is really a *full factor covariance* axis. Experiment 29 above already removes this, and finds the pure-covariance effect larger, not smaller.

A twenty-seed replication of the full factorial (experiment 36) reproduces the ordering. Median misallocation across seeds is 0.232/0.266/0.297/0.343 for plain logit against 0.047/0.077/0.098/0.126 for factor probit over the four strata, and factor probit beats plain logit in all twenty seeds in every stratum. The remaining caveat is the single truth family, skew-normal and close to Gaussian. Figure 3 plots, for each scored single deletion, plain logit’s misallocation against factor probit’s. Points below the diagonal are deletions where factor probit is closer to the truth. Singles and pairs score similarly (factor probit 0.080 versus 0.076 mean misallocated fraction) and are summarized separately in the committed output.

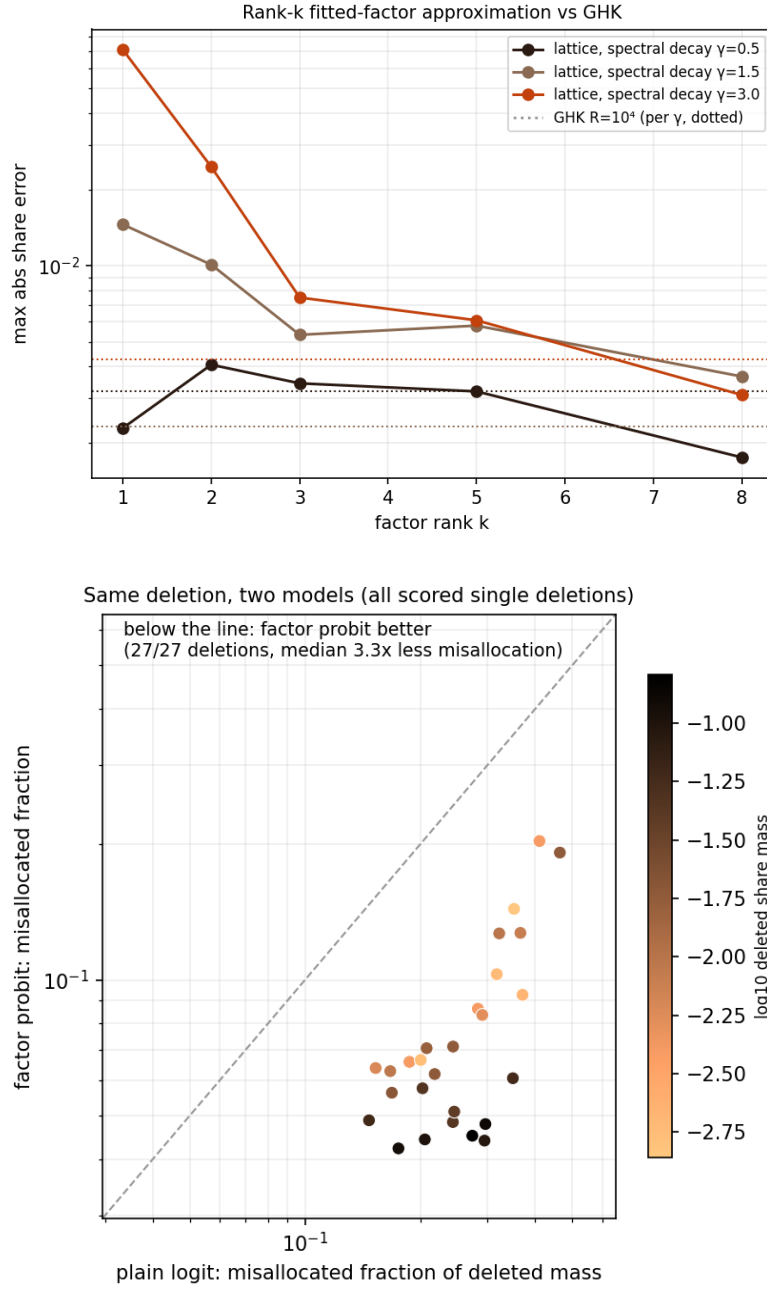


Figure 3: Top: rank- k contrast-space approximation error under spectral decay γ (dotted: GHK at $R = 10^4$, per γ ; not wall-time matched). Bottom: paired comparison on identical deletions—plain logit’s misallocated fraction (horizontal) against factor probit’s (vertical), one point per scored single deletion, shaded by deleted mass; the dashed line is parity.

8 Discussion

Returning to our main limitation of fixed factors, estimation is the natural next step. With the reduced coordinates of Proposition 1, gradients of an outer likelihood or generalized method of moments (GMM) objective pass through the calibration fixed point by implicit differentiation. So there is a route to differentiable probit demand estimation at large N .

Of course, estimating (V, D) rather than supplying them raises the usual identification cautions: scale normalization, factor rotation, and the fact that only difference covariances are identified. Rank and order conditions for identifying error-component structures are given by Walker et al. (2007). This is left to future work.

Separately, the pass itself can be accelerated further. Its two stages are smooth-kernel sums whose matrices are numerically low-rank in our prototype tests, and a Chebyshev-separated evaluation (in the spirit of black-box fast multipole methods) reduces each node’s cost from $O(NL)$ to $O(r(N + L))$, converging quickly in r in those tests. A forward-map prototype is committed (experiment 20). Its systematic error control and extension to the derivative and calibration passes are developed in a companion paper.

Beyond the Gaussian family, the same machinery computes races with arbitrary alternative-specific idiosyncratic densities. Factor probit is best seen as one point in a two-parameter family, base law by factor rank, and every member runs through the same shared-field pass at the same cost. Zero factors recovers the original transform. A common-scale Gumbel base with zero loadings reproduces the Luce model exactly (Luce, 1959; Yellott, 1977), and Gumbel with factors yields correlated softmax.

Neither method class dominates the other. Simulation owns arbitrary Σ and coarse accuracy whereas the shared field owns the factor form, tight accuracy, derivatives, and calibration. Our experiments give some guidance as to the boundary.

In particular, covariance approximation rather than quadrature becomes the binding limitation when the choice-relevant covariance is not well represented at modest rank. We leave to future work a walk-forward calibration on real assortment data. The empirical accuracy evidence for probit against logit families is surveyed in companion work (Cotton, 2026).

9 Potential applications

The territory is larger than econometrics. That is because a probit model is a race in which performances are Gaussian, sometimes correlated. Races take many forms at all levels of spatial and temporal resolution. Predicting item selection and user clicks drives digital commerce, just as predicting market share is fundamental to company analysis. Countless choices are made by man, nature, and machine. Across all the world’s applications of large language models, perhaps a billion next-token choices are made a second.

The engine was canonized independently, under different names, across the quantitative sciences. Thurstone’s comparative judgment, the ordered and binary probits of applied microeconomics, and the Lazear–Rosen tournament (Lazear and Rosen, 1981) are the same Gaussian latent-performance model, and machine learning runs on the same

family: a softmax output layer is its Luce–Gumbel member (Luce, 1959; Yellott, 1977), while Thurstone–Mosteller pairwise comparison (Mosteller, 1951), the probit member, underlies rating systems and preference learning. Bioassay named the probit and made Gaussian tolerances the dose–response standard (Bliss, 1934; Finney, 1947). Quantitative genetics reports the heritability of binary traits on the liability scale, a thresholded latent Gaussian (Dempster and Lerner, 1950; Falconer, 1965). Item response theory began with the normal-ogive curve and adopted the logistic only as a computational stand-in (Lord, 1952). Signal detection’s d' is a double probit (Green and Swets, 1966). The Basel capital formula that regulates the world’s banks is a conditional probit with one common factor (Vasicek, 2002; Gordy, 2003), a rank-one factor model of exactly the form used here. Machine learning made probit its tractable Bayesian classifier (Albert and Chib, 1993) and ships a Thurstonian race as its skill-rating engine (Herbrich et al., 2006). Wherever performance is latent and Gaussian, the same object appears.

Nature repeatedly poses the same computational problem: out of thousands, millions, or vastly more correlated alternatives—candidate crystal-growth directions, microscopic slip systems, adsorption geometries, pore-scale flow routes, turbulent modes, lightning leaders, fault asperities, or phase-transition seeds—which one wins first?

Materials science is especially rich in such races. Any one of a thousand grain boundaries is a competitor at which fracture, diffusion, segregation, or corrosion can initiate first. The same structure appears in precipitation nuclei during alloy aging, hydrogen-trapping sites in embrittlement, electrochemical reaction sites on an electrode, dendrite formation in a battery, charge trapping in semiconductors, and breakdown paths through a dielectric. Biology stages the same contest among mutations and microbial lineages.

Astronomy alone offers several concrete instances. Which of many catalog candidates is the true cross-match, which sky tile hosts a gravitational-wave counterpart, which of a million nightly survey alerts deserves follow-up. Each is a race among thousands of alternatives whose scores share a few instrument and observing-condition factors, and observing a candidate then re-ranking the rest is precisely the removal counterfactual computed here.

It is not only horses that race. However, horse races are worthy of mention, and not only as regards attribution of a core trick used herein. For at least half a century those betting on them have been confronted by the nuances of choice and permutation probabilities in exacta, quinella, trifecta, and place betting. In that setting there has been a strong Darwinian selection that weeds out those anchoring to a simple renormalization of probability—this does not model second-place probabilities well at all if the favorite does not win (Henery, 1981; Lo and Bacon-Shone, 1994).¹³

At the racetrack both nature and the market make choices—the latter in risk-neutral probability. We do not always personify markets as choosers with revealed preferences but perhaps we should. Trillions of dollars are invested passively in capital-weighted indexes in a subtle, unwitting nod to that same simple renormalization deemed unworthy for the

¹³The Darwinian winner is instructive. Benter (1994), whose Hong Kong betting operations by his own concession made close to a billion dollars overall (Chellel, 2018), wrote of the simple renormalization (the Harville formula) that it “is significantly biased, and should not be used for betting purposes.” His fix tempered the win probabilities before conditioning.

racetrack.

Why all this probit work? For anyone modeling performance directly, Gaussian is the natural noise. Sums of many small effects are Gaussian, and nobody measuring times, scores, or energies would posit Gumbel-distributed performance. The Gumbel race is chosen backwards, for the convenience of its winner probabilities. That convenience is starkest when an item is removed from a list whose probabilities have already been assigned.

The reflexive move is the renormalization already met at the racetrack, dividing each survivor by the sum. It feels like no assumption at all, and it is exactly the assumption that performances are independent Gumbel draws. It is also built into the output layer of most neural networks. The honest alternative, recomputing the race under Gaussian performances, has been understandably rare when the race involves a thousand competing choices of token, crystal lattice point, or segment of the sky.

However we hope the main message is not lost. For fifty years, computability has been a very heavy factor in deciding which choice models get used. For the factor form of probit, the form in which correlated utilities entered econometrics (Hausman and Wise, 1978), that constraint is in large part removed by our work. All shares, Jacobian–vector products, and the inverse map remain comfortably below a minute at ten thousand alternatives and rank two on the reported hardware. Integrating factor estimation with this inner solver is, again, outside the present paper, but with (V, D) in hand at modest rank, the choice between logit and factor probit becomes one of fit, not tractability.

A Proof of Proposition 2

We show in order that the Jacobian $J = \partial p / \partial \mu$ is the weighted graph Laplacian with weights w_{ij} (Steps 1–4), that J is positive semidefinite with null space $\text{span}\{\mathbf{1}\}$ (Step 5), and that p restricted to $\mathbf{1}^\perp$ is a smooth bijection onto the interior of the simplex (Steps 6–8). Throughout we work in max-wins coordinates, condition on the factor draw f , write $m_i(f) = \mu_i + v_i^\top f$ and let g_i, F_i be the conditional Gaussian density and CDF, so that

$$p_i = \mathbb{E}_f \int g_i(x | f) \prod_{j \neq i} F_j(x | f) dx.$$

Step 1: off-diagonal entries. Fix $i \neq j$. The mean μ_j enters p_i only through the single factor $F_j(x | f) = \Phi((x - m_j(f))/\sqrt{D_j})$, whose derivative is

$$\frac{\partial}{\partial \mu_j} F_j(x | f) = -g_j(x | f).$$

Differentiation under the integral and the expectation is legitimate because the differentiated integrand is dominated, uniformly in (f, μ) , by an integrable bound:

$$\int g_i(x | f) g_j(x | f) dx = \frac{\exp[-(m_i(f) - m_j(f))^2 / (2(D_i + D_j))]}{\sqrt{2\pi(D_i + D_j)}} \leq \frac{1}{\sqrt{2\pi(D_i + D_j)}},$$

the equality being the Gaussian product integral and the inequality bounding the exponential by one. Hence

$$J_{ij} = \frac{\partial p_i}{\partial \mu_j} = \mathbb{E}_f \int g_i \left(\frac{\partial}{\partial \mu_j} F_j \right) \prod_{\ell \neq i,j} F_\ell dx = -\mathbb{E}_f \int g_i g_j \prod_{\ell \neq i,j} F_\ell dx = -w_{ij}.$$

Step 2: symmetry. The integrand $g_i g_j \prod_{\ell \neq i,j} F_\ell$ is unchanged when i and j are exchanged, so $w_{ij} = w_{ji}$ and $J_{ij} = J_{ji}$.

Step 3: diagonal entries. Translation invariance $p(\mu + c\mathbf{1}) = p(\mu)$ (Proposition 1) makes every row of J sum to zero, $J\mathbf{1} = 0$, so

$$J_{ii} = -\sum_{j \neq i} J_{ij} = \sum_{j \neq i} w_{ij}.$$

The same value follows directly. Since $\partial_{\mu_i} g_i(x | f) = -g'_i(x | f)$, integration by parts gives

$$\frac{\partial p_i}{\partial \mu_i} = \mathbb{E}_f \int (-g'_i) \prod_{j \neq i} F_j dx = \mathbb{E}_f \int g_i \sum_{j \neq i} g_j \prod_{\ell \neq i,j} F_\ell dx = \sum_{j \neq i} w_{ij},$$

the boundary terms vanishing because $g_i \rightarrow 0$ at $\pm\infty$ while $\prod_{j \neq i} F_j$ stays bounded.

Step 4: the Laplacian form. Each rank-one term $w_{ij}(e_i - e_j)(e_i - e_j)^\top$ adds $+w_{ij}$ to the diagonal entries (i, i) and (j, j) and $-w_{ij}$ to the off-diagonal entries (i, j) and (j, i) . Summing over unordered pairs reproduces exactly the entries of Steps 1 and 3, so

$$J = \sum_{i < j} w_{ij}(e_i - e_j)(e_i - e_j)^\top, \quad h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2,$$

the quadratic form following from $h^\top(e_i - e_j) = h_i - h_j$.

Step 5: positive definiteness on the quotient. For every f and finite x the densities g_i, g_j are strictly positive and each $F_\ell(x | f) \in (0, 1)$, so the integrand of w_{ij} is strictly positive and $w_{ij} > 0$. The form $h^\top J h = \sum_{i < j} w_{ij} (h_i - h_j)^2 \geq 0$ then vanishes only when $h_i = h_j$ for every pair, i.e. when h is constant. Thus J is symmetric positive semidefinite with null space exactly $\text{span}\{\mathbf{1}\}$, and $B^\top J B \succ 0$ for every full-rank basis B of $\mathbf{1}^\perp$. The identity $p = \nabla G$ for $G(\mu) = \mathbb{E} \max_i \{\mu_i + \xi_i\}$ is the Williams–Daly–Zachary/McFadden identity (Williams, 1977; Daly and Zachary, 1978; McFadden, 1981), so $J = \nabla^2 G$ is the Hessian of the convex G .

Step 6: existence of the inverse. Fix a strictly positive target $\hat{p}$ in the interior of the simplex and minimize

$$H(\mu) = G(\mu) - \hat{p}^\top \mu \quad \text{over } \mu \in \mathbf{1}^\perp.$$

On $\mathbf{1}^\perp$, H is strictly convex, its Hessian being the reduced $J \succ 0$ of Step 5. Since the shocks are centered, Jensen's inequality gives $G(\mu) = \mathbb{E} \max_i (\mu_i + \xi_i) \geq \max_i (\mu_i + \mathbb{E} \xi_i) = \max_i \mu_i$. Using $\mathbf{1}^\top \hat{p} = 1$,

$$\begin{aligned} H(\mu) &\geq \max_i \mu_i - \hat{p}^\top \mu = \sum_i \hat{p}_i (\max_j \mu_j - \mu_i) \\ &\geq \hat{p}_{\min} \sum_i (\max_j \mu_j - \mu_i) \geq \hat{p}_{\min} (\max_i \mu_i - \min_i \mu_i) \\ &\geq \hat{p}_{\min} \|\mu\|_\infty \geq \hat{p}_{\min} \|\mu\|_2 / \sqrt{N}. \end{aligned}$$

The second line uses $\hat{p}_i \geq \hat{p}_{\min} > 0$ and $\max_j \mu_j - \mu_i \geq 0$; the third keeps only the single term $i = \arg \min_i \mu_i$; and the fourth uses that a mean-zero vector has $\max_i \mu_i \geq 0 \geq \min_i \mu_i$, so its range dominates $\|\mu\|_\infty$. Hence $H(\mu) \rightarrow +\infty$ as $\|\mu\|_2 \rightarrow \infty$ on $\mathbf{1}^\perp$: H is coercive and attains its minimum.

Step 7: the minimizer inverts p . Because $D_i > 0$, the utilities have a smooth full-support joint density and ties are null, so $\nabla G = p$. At the minimizer μ^* the gradient of H vanishes on the constraint subspace, $B^\top(p(\mu^*) - \hat{p}) = 0$, so $p(\mu^*) - \hat{p} \in \text{span}\{\mathbf{1}\}$. Both $p(\mu^*)$ and $\hat{p}$ sum to one, so the constant is zero and $p(\mu^*) = \hat{p}$. Strict convexity makes μ^* the unique mean-zero solution.

Step 8: smoothness. The map p is smooth and its reduced Jacobian $B^\top J B$ is non-singular everywhere on $\mathbf{1}^\perp$ (Step 5), so the inverse-function theorem makes $\hat{p} \mapsto \mu^*(\hat{p})$ smooth on the interior of the simplex. This completes the proof of Proposition 2.

B Derivation of the conjugate trace identity

Work in max-wins coordinates at fixed mean-zero μ , with $\xi \sim N(0, \Sigma)$, density φ , winner $I = \arg \max_i (\mu_i + \xi_i)$, and choice cells $A_i = \{x : \mu_i + x_i \geq \mu_j + x_j \ \forall j\}$.

Step 1: reduce to the winner's mean shock. $G(\mu) = \mathbb{E}[\mu_I + \xi_I] = \mu^\top p + \mathbb{E}[\xi_I]$, so the claim is $\mathbb{E}[\xi_I] = \text{tr}(\Sigma J)$.

Step 2: integrate by parts on each cell. Since $\nabla \varphi(x) = -\Sigma^{-1} x \varphi(x)$, we have $x \varphi(x) = -\Sigma \nabla \varphi(x)$, and the divergence theorem applied coordinatewise on the cell (Gaussian decay kills the boundary at infinity) gives

$$\mathbb{E}[\xi \mathbf{1}\{\xi \in A_i\}] = \int_{A_i} x \varphi(x) dx = -\Sigma \oint_{\partial A_i} \varphi(x) n(x) dS(x),$$

with n the outward unit normal and dS surface measure.

Step 3: identify facet masses with photo-finish weights. The boundary of A_i is the union of facets $F_{ij} = \{x : \mu_i + x_i = \mu_j + x_j \geq \mu_\ell + x_\ell \ \forall \ell\}$ with outward normal $n = (e_j - e_i)/\sqrt{2}$. The facet mass is the photo-finish weight, $w_{ij} = \frac{1}{\sqrt{2}} \int_{F_{ij}} \varphi dS$, the surface form of the weights of Proposition 2 stated in the transport discussion. Hence

$$\oint_{\partial A_i} \varphi n dS = \sum_{j \neq i} \frac{e_j - e_i}{\sqrt{2}} \int_{F_{ij}} \varphi dS = \sum_{j \neq i} (e_j - e_i) w_{ij},$$

and therefore $\mathbb{E}[\xi \mathbf{1}\{I = i\}] = \Sigma \sum_{j \neq i} w_{ij} (e_i - e_j)$. Its i th coordinate is $(\Sigma J)_{ii}$, since $J e_i = \sum_{j \neq i} w_{ij} (e_i - e_j)$.

Step 4: sum over cells. Using $w_{ij} = w_{ji}$ and symmetrizing,

$$\mathbb{E}[\xi_I] = \sum_i e_i^\top \Sigma \sum_{j \neq i} w_{ij} (e_i - e_j) = \sum_{i < j} w_{ij} (e_i - e_j)^\top \Sigma (e_i - e_j) = \text{tr}(\Sigma J),$$

and $(e_i - e_j)^\top \Sigma (e_i - e_j) = D_i + D_j + \|v_i - v_j\|^2 = s_{ij}^2$. Finally $\Omega(p) = p^\top \mu - G(\mu) = -\text{tr}(\Sigma J)$ at $p = p(\mu)$. Experiment 27 verifies the per-class identity $\mathbb{E}[\xi_i \mathbf{1}\{I = i\}] = (\Sigma J)_{ii}$ within Monte Carlo noise, the full identity to 3×10^{-9} , and the binary closed form $\Omega = -s \phi(\Phi^{-1}(q))$ to 3×10^{-13} .

Reproducibility and Supplementary Materials

Code and experiments: all numbers come from committed, seeded scripts with correctness anchors run before any comparison (repository experiments 13–27, 29–31, and 33–38, under `research/experiments/`; experiment 28, which supports the companion paper, and exploratory experiment 32 remain in the `kinetics` research repository). A single manifest, `research/experiments/run_all_paper.py`, regenerates every table and figure; this paper is built from release `v1.1.0` of the repository at <https://github.com/microprediction/winning>, whose package is on PyPI (`pip install winning`).

Software: the algorithm ships in the open-source `winning` package (Cotton, 2021b), module `winning.factor`, with an optional compiled Rust kernel (`fastrace`); package–benchmark agreement is itself a reported number.

Data: all experiments use synthetic data generated by the seeded scripts; no external data are required.

Hardware and software: Apple M4, 16 GB, Python 3.12.9, NumPy 2.4.6, SciPy 1.18.0; compiled timings use `rustc` 1.97.1 (release profile, rayon defaulting to all performance cores); experiment 17 enforces single-threaded BLAS via `threadpoolctl`; sub-second timings are warmed medians of three runs, and long simulations are single realizations, disclosed as such. Monte Carlo chunk sizes derive from an explicit 1.5 GB memory budget, and peak resident memory stays below 2 GB.

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